Supplementary material

Cryptic genomic variation in a panmictic squid: an ancient inversion supergene and genome-wide signatures of positive selection in *Illex illecebrosus*

November 13, 2026

Supplementary methods

Read processing and variant calling. Raw reads for all 633 individuals were downloaded from ENA project PRJEB101095 and processed with grenepipe (Czech and Exposito-Alonso, 2022). Twelve alignments hold almost no reads (93 to 2,009,803 autosomal reads in `samtools idxstats`, against a median of 57,182,056, listed in `data/qc/empty_bams_12.tsv`), eleven from division 4X and one from 3K. They were excluded from every count and estimate. Analyses that were run on all 633 received only missing genotypes or no reads from them, so those results rest on the remaining 621. Two further individuals, SQ23HBB023 (2+/2+, division 6C) and SQ23HBB124 (2a/2a, 6A), were excluded afterwards as likely contaminated (`data/qc/excluded_14.tsv`). Both had been called female from Z coverage yet carry Z heterozygosity in the male range (0.0096 and 0.0036, against a female 95th percentile of 0.0014), Z coverage above every male (1.46 and 1.61, against a male maximum of 1.34), and the two most extreme scores on the second autosomal principal component. Counts, karyotype frequencies, the principal-component analyses, the karyotype-by-division and capture-depth tests are for the remaining 619. Genome-wide and inversion analyses run before the exclusion, including the spectra, demography, sweep scans, recombination maps, geographic F_{ST} scans and the inversion statistics, include the two (2 of 621 individuals). Reads were trimmed with fastp (Chen et al., 2018) and mapped to the `Illex_F24` primary assembly with BWA-MEM2 (Vasimuddin et al., 2019), and duplicates were removed with Picard MarkDuplicates. Each individual was genotyped with GATK HaplotypeCaller in gVCF mode, and the gVCFs were jointly genotyped with all sites retained (Poplin et al., 2017). Variant records were annotated with snpEff and soft-filtered with bcftools (Danecek et al., 2021). A site was removed if its quality was below 20, its root-mean-square mapping quality was below 40, or its strand-bias score (FS) exceeded 60. It was also removed if it was missing in half or more of the individuals, if it lay in a GC-corrected high-depth window identified with mosdepth, or if it lay outside the accessibility mask. Sites at intermediate allele frequency (0.1 to 0.9) with a strong excess of heterozygotes (ExcHet < 0.001) were flagged as likely paralogous and removed. Only sites passing every filter were used. The accessibility mask is the 1.43 Gb of the 3.47-Gb assembly that lies outside repeats annotated with Earl Grey. Mappability was computed with GenMap 1.3.0 (Pockrandt et al., 2020) for 150-mers with up to two mismatches, matching the 151-bp reads. Across the assembly 14.7% of positions are not uniquely mappable, but within the accessibility mask only 3.8% (53.7 Mb) are, because the repeat annotation already removes nearly all multi-copy sequence. Non-unique sequence is

concentrated six-fold in the 22 anomalous-diversity blocks (23% of their accessible sequence), and outside them it is 2.7% of accessible sequence. For the sweep scan and the gene-set analyses, the mask was extended by the 22 anomalous-diversity blocks (129 Mb, Supplementary Table S9), in which collapsed duplications and sequence duplicated within the assembly make genotype-based statistics unreliable.

Diversity and demography. All site-frequency spectra and diversity statistics were computed with ANGSD 0.940 (Korneliussen et al., 2014) directly from the deduplicated BAM alignments of the 621 individuals with data, not from the called genotypes, so that the uncertainty of genotypes at about 9-fold depth is carried through as genotype likelihoods. Each chromosome was run separately with the options `-doSaf 1 -doCounts 1 -GL 1 -minQ 20 -minMapQ 20 -remove_bads 1 -only_proper_pairs 1 -minInd 1 -setMinDepthInd 1`, restricted with `-sites` to the accessibility mask and with the reference supplied as both `-ref` and `-anc`. `-GL 1` is the SAMtools genotype-likelihood model, and because the reference is not an ancestral sequence all spectra are folded. The folded spectrum was estimated with `realSFS -fold 1 -tole 1e-8 -maxIter 200`, per-site thetas with `realSFS saf2theta -fold 1`, and π, Watterson's θ and Tajima's D with `thetaStat do_stat`, per chromosome and in 50-kb windows (10-kb step genome-wide, 50-kb step per division). The genome-wide folded spectrum pooled across chromosomes is the input to the demographic models. Per-division statistics used the same options on at most 25 individuals per division (all individuals in 3O, 4S and 4T), and the Z was run in the 329 confidently sexed males with data. Per-karyotype spectra and genotype-likelihood F_{ST} between arrangements used the 2a/2a and 2+/2+ homozygotes as separate sample sets. Stairway Plot 2, moments (a two-epoch growth model fitted to the folded spectrum, with Godambe confidence intervals), and GONE2 were fitted as described in Supplementary Table S1, and the moments model was checked by forward simulation, which reproduced the observed π and Tajima's D.

Genotype-based statistics. Statistics that need individual genotypes were computed with pg_gpu (Pope et al., 2026) from the filtered VCF. Windowed π, d_{xy} and Hudson F_{ST} between arrangements used biallelic SNPs in 100-kb windows, with the 2a/2a ($n = 223$) and 2+/2+ ($n = 97$) homozygotes as the two populations and missing genotypes included pairwise. The between-division F_{ST} matrix (Hudson), the linkage-disequilibrium triangles (genotype r^2 among common SNPs), and the principal-component analyses were computed the same way. The per-SNP geographic F_{ST} scans (Supplementary Figs. S2, S19) used plink2 (Chang et al., 2015) to filter and count alleles at every biallelic SNP with a minor-allele frequency above 0.05 in the pooled 620 individuals from the ten divisions with more than one sample (division 3N, with a single individual, was excluded). Per-division allele counts were converted to Hudson F_{ST} for each of the 45 division pairs (Bhatia et al., 2013), at sites genotyped in at least 40 individuals in that pair and with at least two alleles in each division. The per-SNP value is the ratio of averages over informative pairs (the sum of the numerators over the sum of the denominators), without clipping, keeping sites informative in at least 20 of the 45 pairs. Of the 23,892,020 SNPs passing the frequency filter, 20,276,930 met this criterion, 20,117,984 of them on the 45 autosomes (133,957 in the chromosome-2 inversion) and 158,946 on the Z chromosome. The null was built by shuffling division labels among the individuals 20 times (division sizes kept) and recomputing F_{ST} at the same SNPs in the same way. Thresholds are quantiles of the pooled null over the autosomes outside the chromosome-2 inversion (3.996×10^8 values), and the expected count above each threshold is the number of SNPs times the null tail rate.

The inversion. Karyotypes were called from the first principal component of a local PCA across chromosome 2 with a three-component Gaussian mixture. To control for sequencing depth, every alignment was downsampled to about 1× in chr2:60–80 Mb (samtools, fixed seed), re-genotyped, and karyotyped again in the same way. The two assignments agree for 601 of the 619 individuals (97.1%). All 18 disagreements are full-depth 2a/2a calls, 15 of them 2a/2+ and 3 of them 2+/2+ at equal depth. Twelve have low read depth (7–22 million autosomal reads, against a median of 57 million), so their full-depth genotypes carry little heterozygosity and their first principal component is drawn toward the centre, where the broad 2a/2a mixture component absorbed them. The other six have heterozygosity in the inversion typical of 2a/2+ (0.05–0.09). The final calls take the equal-depth assignment for these 18 (`data/qc/karyotypes_619.tsv`), giving 223 2a/2a, 299 2a/2+ and 97 2+/2+. Breakpoints were placed at the step change in between-arrangement differentiation and confirmed by a strand flip in the AnchorWave alignment to *I. coindetii*. The arrangement carried by each haplotype of the reference individual was read from minimap2 2.29 (`-x asm5`) alignments of chromosome 2 of the two haplotype assemblies to each other and to the primary assembly, and the orientation of each haplotype relative to *I. coindetii* from alignments to the homologous *I. coindetii* chromosome (OZ346549.1, `-x asm20`). Alignment blocks of at least 1 Mb were ordered along each *Illex illecebrosus* haplotype, and polarity was read from which arrangement keeps the order and adjacency of *I. coindetii* sequence across the rearranged segment (Supplementary Table S3). The same comparison was made with the chromosome-level assembly of *Sthenoteuthis oualaniensis* (Li et al., 2022), a member of the same family. Its chromosome-2 homolog, HiC_scaffold_1, is the scaffold that receives 265 of the 319 *Illex illecebrosus* chromosome-2 genes transferred with LiftOn (Chao et al., 2025), and chromosome 2 of each haplotype assembly was aligned to it with minimap2 (`-x asm20`). Because the two genera differ by many small rearrangements along the chromosome, polarity was read at each breakpoint from which flanking segments are joined, using alignments of at least 5 kb with mapping quality of at least 5 together with the order of the transferred genes. Allele sharing with *I. coindetii* was also scored at the 15,093 sites in the inversion where allele frequency differs by at least 0.7 between 2a/2a and 2+/2+ homozygotes, with the *I. coindetii* base read from the AnchorWave alignment of *I. coindetii* to the primary assembly, which carries 2+ in the inversion. Where the *I. coindetii* base was callable (14,650 sites), the majority allele of 2a matches it at 26.0% of sites, so 2+ carries the *I. coindetii* allele at most diagnostic sites. The 2a matches are consistent with variation that was segregating in the ancestral population and was captured by the inversion when it formed, possibly in more than one capture event, which would give the soft start discussed in the main text. Chromosome 2 of both outgroups is rearranged in this region, so the arrangement of the common ancestor away from the breakpoints cannot be determined. It may have been an intermediate rearranged form, and the polarity used here is the reading on which *I. coindetii* and *S. oualaniensis* agree at both breakpoints. Restricted to the 4,582 sites where *I. coindetii* differs from the reference base, 2a carries the *I. coindetii* allele at 65.0% (2,980) and 2+ at 35.0% (1,602), but this subset is selected for sites at which the reference, a 2+ haplotype, carries a non-*I. coindetii* allele, so it is biased toward 2a matches. Reads of ten *I. argentinus* individuals (about 0.3× each) mapped to the primary assembly were counted at the same diagnostic sites (`bcftools mpileup`, mapping and base quality of at least 20). Reference bias was measured at 150,000 SNPs in the collinear control (10–30 Mb), where the *I. argentinus* non-reference read fraction tracks the *Illex illecebrosus* non-reference allele frequency p as $0.018 + 1.02p$, and the frequency of 2a in *I. argentinus* was estimated by maximum likelihood from the read counts under this calibration. All ten individuals give the same estimate (0.27–0.30, pooled 0.284), rather than the values near 0, 0.5 and 1 expected if the inversion segregated in *I. argentinus*. At sites covered by at least four reads, 87% carry reads of only one allele, as expected if each site is fixed, so *I. argentinus* carries the 2+ allele at most diagnostic sites and the 2a allele

at the rest. Its 2a alleles fall mainly at sites where *I. coindetii* also carries the 2a allele (0.68 of reads, against 0.24 where *I. coindetii* carries the 2+ allele), so they are best read as ancestral states shared by both species, and the inversion polymorphism is not shared with *I. argentinus*. The age and frequency trajectory were fitted with msinv as described in the next paragraphs. Unannotated, non-repeat sequence in the inversion was searched with DIAMOND `blastx` against seven cephalopod proteomes, with transposable elements from Earl Grey (Baril et al., 2024) and transcripts from StringTie2 (Kovaka et al., 2019) as supporting evidence. This recovered 14 likely protein-coding loci that the annotation lacks. They are provided as `protein_match` features in version 3 of the annotation and are not counted among the 89 gene models of the inversion. Fixed differences were sites with an allele-frequency difference of at least 0.9 between 2a/2a and 2+/2+ homozygotes. Gene-ontology enrichment of the inversion genes used Fisher's exact test against the annotated-gene background in all three ontologies, compared with 500 random gene sets of equal size, and gene density was compared with all autosomal windows of the same length (0.5-Mb steps). The zero-fold fraction among the top 1% of coding sites by per-site F_{ST} (genotype likelihoods, 40 individuals per karyotype) was compared with all coding sites of the same region by Fisher's exact test, in the inversion ($p = 1 \times 10^{-32}$) and in collinear chr2:10–30 Mb ($p = 0.007$). Per-transcript d_{xy}, the mean between-arrangement divergence over a transcript's coding sites (transcripts with at least 50), was normalised to its 100-kb window and compared with the same ratio for collinear control transcripts.

Inversion coalescent model. The inversion was simulated with msinv v0.4.1 (`https://github.com/stsmall/msinv`), a coalescent simulator with recombination that writes tree sequences. It uses the sequentially Markov approximation (McVean and Cardin, 2005), extended to two karyotype classes as in Peischl et al. (2013), in which two lineages coalesce only where they share ancestral material. The model code, inference scripts and every result quoted here are in the `illex/` and `results/illex/` directories of that repository (branch `illex/baseline-refit-karyofix`). Inside the inversion every lineage carries a karyotype label, ancestral (S, the 2+ arrangement) or derived (I, the 2a arrangement). Two lineages of the same class coalesce at rate $1/[2N(t)\,p_c(t)]$ per generation, where $N(t)$ is the effective size and $p_c(t)$ the frequency of their class, and a lineage of class c recombines within the inversion at rate $r\,p_c(t)$, because crossing over occurs only between chromosomes of the same arrangement. Lineages of different classes cannot coalesce while the barrier exists, except through gene flux, which copies tracts of mean length 300 bp between arrangements, within the 188–390 bp estimated by Korunes and Noor (2019), with the position-dependent initiation rate of Peischl et al. (2013), rising from zero at the breakpoints toward the centre of the inversion. The gene-flux parameter is the per-site conversion probability per generation in heterokaryotypes. Backward in time, the barrier is removed at t_{inv}, when the derived lineages rejoin the ancestral class. Each proposal simulated 40 kb of inversion between two 10 kb collinear margins, and a separate 40 kb of collinear sequence with msprime 1.4.1 (Baumdicker et al., 2022) as the control, with 100 chromosomes of each arrangement and $\mu = 3 \times 10^{-9}$. For linkage disequilibrium the chromosomes of each arrangement were paired at random into 50 diploids and passed through the genotype error model of the data, in which a heterozygote is called homozygous with probability h and each genotype is missing with probability one minus the call rate, both estimated for each arrangement and region from the same individuals as the observed statistics (call rates 0.62–0.67, $h = 0.64$–0.69).

Neutral baseline. The demography and recombination rate of the simulations were fitted to collinear chromosome-2 sequence flanking the inversion, chr2:40–55 and 85–115 Mb, by rejection

ABC. This sequence lies 5.5–35.5 Mb outside the breakpoints and avoids a stretch of doubled diversity at chr2:24.1–35.9 Mb. The 2a/2a and 2+/2+ individuals were treated as two random sample sets, and the fitted statistics were the folded site-frequency spectrum in six bins, linkage disequilibrium in six distance bins, Tajima's D and $\log_{10}\pi$, each averaged over the two sample sets, together with eight cells of the joint spectrum between them (the four cells for variants near fixation in one set and absent from the other are empty in collinear sequence and were left out). Diversity in this sequence, from the summed within-group ANGSD spectra, is $\pi = 0.0061$, against 0.0093 for the autosomes. The model was exponential growth from N_{anc} to N_0 over T_{grow} generations, constant before that, with recombination rate r and μ fixed at 3×10^{-9}. 20,000 draws from log-uniform priors (N_{anc} 10^5–3×10^6, growth fold 1–200, T_{grow} 10^4–3×10^6, r 10^{-10}–10^{-8}) were each simulated for 200 kb with msprime and the control's genotype error model. The 1% closest to the data in a MAD-scaled Euclidean distance were accepted with Epanechnikov weights, and the log parameters were adjusted by local-linear regression (Beaumont et al., 2002). The posterior medians (95% intervals) are N_{anc} = 507,288 (378,143–635,672), N_0 = 9,112,696 (7,632,032–11,683,078), T_{grow} = 382,957 (314,983–472,238) and $r = 5.51\times10^{-10}$ (4.61–6.49×10^{-10}). With μ fixed, only r/μ is identified, so r is the effective rate for these statistics. The accepted draws reproduce the spectrum, Tajima's D, π, linkage disequilibrium from 240 bp to 20 kb, and the symmetric cells of the joint spectrum. Linkage disequilibrium at 100–240 bp lies above every accepted draw, and 2+/2+ carries more private rare variants than 2a/2a (0.352 against 0.272 of variable sites), which a single panmictic population does not produce. The inversion model used the posterior medians.

Trajectory models and priors. The frequency of 2a followed a Wright–Fisher diffusion in steps of 10 generations, with the effective size of the neutral baseline. Under neutral drift (N) there is no selection. Under directional selection the frequency changes deterministically by $s\,p(1-p)$ per generation from t_{sel} to the present (D1) or only until t_{stop}, with drift alone after that (D2). Under balancing selection (B) it changes by $s\,p(1-p)(p^*-p)/p^*$ from t_{sel}, the form expected under overdominance with equilibrium p^*. A hard start places a single copy of 2a at t_{inv}, with selection from t_{inv}, and its path is weighted by the exact probability that the copy establishes. A soft start places 2a at a founding frequency p_0 at t_{inv}, with neutral drift until t_{sel}. A path is kept only if 2a is neither lost nor fixed and ends within 0.02 of its present frequency of 0.602. The priors were t_{inv} uniform on 10^5–1.5×10^6 generations, p_0 log-uniform on 10^{-3}–0.10, $t_{\mathrm{sel}} = f\,t_{\mathrm{inv}}$ and $t_{\mathrm{stop}} = f'\,t_{\mathrm{sel}}$ with f and f' uniform on 0–1, s log-uniform on 10^{-6}–10^{-1}, p^* uniform on 0.3–0.95, the gene-flux rate log-uniform on 10^{-9}–10^{-5} when gene flux was fitted and 10^{-15} otherwise, and equal prior probabilities of a hard and a soft start. The upper bound of 0.10 on p_0 is the largest founding frequency we consider biologically reasonable.

Summary statistics. Thirty-nine statistics were fitted. Three are ratios between arrangements, π_I/π_S, d_{xy}/π_I and the ratio of their singleton-class fractions. Twelve are the folded site-frequency spectra of the two arrangements projected to 20 chromosomes, as proportions of segregating sites in bins 1 to 5 and 6 or more. Twelve are the mean genotype r^2 between SNPs with a minor-allele frequency of at least 0.05 within each arrangement, in six log-spaced distance bins from 100 bp to 20 kb. Twelve are the cells of the folded joint spectrum, with both arrangements projected to 20 chromosomes and binned at 0, 1–4, 5–15, 16–19 and 20 derived copies. Every statistic except the joint spectrum is divided by the same statistic in the control, in the simulations and in the data alike. In the data the ratios come from called genotypes of the 223 2a/2a and 97 2+/2+ individuals (pg_gpu), the spectra from per-arrangement ANGSD site-frequency spectra, the joint spectrum from a two-dimensional realSFS spectrum of 40 individuals of each homokaryotype, and linkage

disequilibrium from 20 random subsamples of 50 individuals of each arrangement. The control is the collinear sequence of the neutral baseline, and its joint spectrum was built with the same 80 individuals, sites and filters as the inversion's. Three further statistics were withheld from the fit and used only to check it, the Tajima's D of each arrangement and the ratio of d_{xy}/π in windows near the breakpoints to that in windows at the centre of the inversion.

ABC random-forest model choice. The four histories were compared by ABC-RF (Pudlo et al., 2016) in abcrf, on a reference table simulated from the priors of the four histories with gene flux, with hard and soft starts simulated separately and each path kept when it passed the present-frequency filter (16,959 simulations after 19 with undefined statistics were removed, 3,228 for N, 3,744 for D1, 4,994 for D2 and 4,993 for B). Hard starts were weighted by their establishment probability, and each start class by its prior probability and its acceptance rate, so that each history carries its prior-weighted mass. A forest of 1,000 trees was grown on the 39 statistics and their linear discriminant axes, and the posterior probability of each history is the prior-weighted mass of the training simulations that share leaves with the observed statistics, multiplied by the probability that a prior draw of the history passes the present-frequency filter (0.0064 for B and 0.0005–0.0006 for each of the others). At the observed statistics the forest alone gives 0.31 for D2, 0.29 for B, 0.24 for N and 0.17 for D1, and with the acceptance rates the posterior probabilities are 0.83 for B, 0.08 for D2, 0.06 for N and 0.04 for D1. The classifier was assessed by grouped five-fold cross-validation (Supplementary Fig. S26). Its accuracy is 0.51 against 0.25 by chance. Neutral drift and ongoing directional selection are recovered best (0.86 and 0.68), whereas balancing selection is recovered in 0.36 of its mass and confused mainly with neutral drift (0.33), and a stalled sweep is recovered in 0.14 and confused with the other three.

Simulated length. Each simulation covers 40 kb of inversion, against 18.96 Mb in the data. To test whether this length represents the whole inversion, 48 NPE posterior draws of the balancing model with gene flux were each given one frequency path, and the same path was simulated over 40 kb, 160 kb, 640 kb, 2.56 Mb and the full 18.96 Mb. For the 47 draws with a valid path (one draw produced none within 200 attempts), the mean of every fitted and withheld statistic at 18.96 Mb lies within 0.2 standard deviations (of the 40-kb values across draws) of its mean at 40 kb. The spread across draws is smaller at 18.96 Mb, 0.3–1.0 of that at 40 kb for the fitted statistics and 0.12 for the edge-to-centre ratio. The 40-kb simulations therefore reproduce the expected statistics of the full inversion with more variance, which widens the posterior rather than shifting it.

Neural posterior estimation. Parameters of the balancing model were estimated by neural posterior estimation (Cranmer et al., 2020; Min et al., 2026) in sbi 0.27 (Tejero-Cantero et al., 2020; Boelts et al., 2025). The training simulations were new draws from the prior of the balancing model with a founding frequency, with and without gene flux, kept when the present frequency of 2a fell within 0.02 of 0.602, with the same simulator, baseline and 39 statistics (59,743 simulations with gene flux and 59,531 without, each from about 5.4 million prior draws). Starts from a single copy, which carry 3% of the posterior mass of balancing selection in the ABC-RF reference table, were not included. The parameters were transformed to the scale on which their priors are uniform (t_{inv}, t_{sel}/t_{inv}, $\log_{10} p_0$, $\log_{10} s$, p^* and the log gene-flux rate), and the statistics were standardised. The posterior is an equal mixture of five neural spline flows (Durkan et al., 2019) with five transforms and 64 hidden units, each trained from a different seed with 10% of the training set held for validation and stopped after 30 epochs without improvement. Calibration was tested on 2,000 held-out simulations by simulation-based calibration (Talts et al., 2018) and by the TARP coverage

test (Lemos et al., 2023). With gene flux the central 50, 90 and 95% intervals contain the true value in 49–52, 89–91 and 94–96% of simulations, and the rank distributions are uniform (Kolmogorov–Smirnov p 0.12–0.68) except for a slight deviation for p^* ($p = 0.02$). The observed statistics lie inside the training distribution, at the 0.27 quantile of the robust Mahalanobis distance and between the 29th and 89th percentile for each statistic. Without gene flux the coverage is also nominal and the ranks are uniform (p 0.09–0.64), but about half of the training simulations have no variants shared at low frequency by the two arrangements, which the data have. Twenty thousand posterior draws were taken at the observed statistics. The trajectories in main-text Fig. 4b and Supplementary Fig. S14 were simulated from 400 posterior draws with gene flux, each re-simulated until its path passed the present-frequency filter (392 within 50 attempts).

Posterior predictive checks. The balancing model was checked from 1,000 NPE posterior draws for each gene-flux setting. Each draw was simulated again with the same simulator, with the frequency path retried until it passed the present-frequency filter. Each statistic was scored by the quantile of the observed value among the replicates and a two-sided predictive p-value, and the joint fit by the rank of the observed distance from the predictive median, in the distance of the fit, among the replicates' distances (Supplementary Fig. S25 for the balancing model with gene flux). Under balancing selection with gene flux the joint predictive p is 0.51 (0.09 without gene flux). The withheld statistics are reproduced (two-sided p 0.84 and 0.56 for Tajima's D in 2a and 2+, 0.99 for the edge-to-centre ratio). With gene flux, no statistic falls below $p = 0.05$, and the cell of variants at low frequency in both arrangements, linkage disequilibrium at 8–20 kb in 2+ and at 100–580 bp in 2a, and the cell of variants rare in 2a and fixed in 2+ are at the edge of their ranges ($p = 0.07$–0.14). Without gene flux, no statistic falls below $p = 0.05$ either, and linkage disequilibrium at 8–20 kb in 2+ ($p = 0.07$) and the cells of variants rare in 2a and at high frequency or fixed in 2+ are at the edge of their ranges (observed 0.016 and 0.011, predicted 0.002 and 0.0002, $p = 0.12$ and 0.06). The cell of variants at low frequency in both arrangements is reproduced in both settings (observed 0.064, predicted 0.031 with gene flux and 0.035 without, $p = 0.07$ and 0.29). In the collinear flanks the same cell matches the neutral baseline (observed 0.233, accepted draws 0.210–0.269), so the two-dimensional spectrum and the baseline reproduce it where there is no barrier.

Recombination map. ReLERNN (Adrion et al., 2020) was run in unphased mode on biallelic SNPs with at most 50% missing genotypes. Individuals with no genotype calls on a chromosome cannot enter the unphased model, and the 329 confidently sexed males with data were used. The autosomal network was trained on chromosomes 1, 13 and 45 with 12,000 coalescent simulations (with 1,000 each for validation and testing) under the Stairway Plot 2 demography with $\mu = 3 \times 10^{-9}$, an upper bound of 4 on the ratio of ρ to θ, and 6-kb windows, masked by the inaccessible regions and the 22 π-elevated blocks with a window-level masking threshold of 0.9. It was trained for 50 epochs and applied to every autosome except chromosome 2, in windows of 3–6 kb. The Z network was trained in the same way in 30-kb windows, with the Z accessibility mask and the effective sizes of the demography multiplied by 0.60, the Z-to-autosome diversity ratio. Predictions were corrected with 200 parametric-bootstrap replicates in each of 20 rate slices. The map pools all karyotypes and reads linkage disequilibrium within windows of a few kilobases, so it does not resolve the barrier between chromosome-2 arrangements, which is shown instead by long-range linkage disequilibrium (Supplementary Figs. S4, S24).

Z recombination and sex-biased mutation. Linkage disequilibrium on the Z and on the autosomes was compared in the 329 ZZ males, the only individuals that carry two Z copies. Two-locus statistics (D^2 and π_2, unbiased for unphased genotypes (Ragsdale and Gravel, 2020)) were computed with pg_gpu (Pope et al., 2026) on accessible biallelic SNPs in 2-Mb windows, 86 on the autosomes (at 10 and 40 Mb of each autosome, excluding the chromosome-2 inversion and the anomalous-diversity blocks) and nine spaced every 4 Mb along the Z, with SNP pairs binned by distance from 0 to 5 kb. The decay of $\sigma_D^2 = D^2/\pi_2$ with distance was fitted to the expectation of Ohta and Kimura (1971) plus a constant sample-size term, giving the population-scaled rate $\rho = 4N_e r$ per base for each chromosome class. The ratio ρ_Z/ρ_A was also estimated without a model, as the distance rescaling that best superimposes the two decay curves. Confidence intervals come from 200 bootstrap resamples of windows. Because the Z recombines only in males and spends two thirds of its time in them, $\rho_Z/\rho_A = (N_{e,Z}/N_{e,A})\,(4/3)\,r_m/(r_m + r_f)$, where r_m and r_f are the male and female per-generation rates and the male rate is taken to be the same on the Z and the autosomes. With $N_{e,Z}/N_{e,A} = 3/4$, equal recombination in the two sexes predicts $\rho_Z/\rho_A = 0.5$ and achiasmate female meiosis predicts 1 (Coffing et al., 2026).

Sex-biased mutation was estimated from synonymous divergence between *Illex illecebrosus* and *S. oualaniensis*, whose chromosome 46 is syntenic with the *Illex illecebrosus* Z (Small et al., 2026). The *Illex illecebrosus* annotation was transferred to the *S. oualaniensis* assembly with LiftOn (Chao et al., 2025), one transcript per gene was aligned codon by codon with MACSE 2.07 (Ranwez et al., 2018), which allows frameshifts in the transferred sequence, and codons with gaps, frameshifts, ambiguous bases or stop codons were removed. Synonymous and nonsynonymous divergence (d_S, d_N) were computed per gene by the method of Nei and Gojobori (1986) with a Jukes–Cantor correction. Genes in the chromosome-2 inversion or the anomalous-diversity blocks were excluded, leaving 8,649 autosomal and 109 Z-linked genes with $d_S > 0$. The male-to-female mutation ratio $\alpha = \mu_m/\mu_f$ was obtained from the ratio R of mean Z to mean autosomal d_S as $\alpha = (3R-2)/(4-3R)$, following Miyata et al. (1987), as in Coffing et al. (2025) and Coffing et al. (2026). Z-linked genes carry more gene-body CpG methylation than autosomal genes in *Illex illecebrosus*, most strongly in females (Small et al., 2026). If this extends to the germline, methylation-associated mutation raises the Z rate and biases α upward, so the female bias is if anything underestimated. To separate mutation from selective constraint and base composition, R was recomputed with autosomal genes matched to the Z genes by deciles of d_N/d_S and of third-position GC content (GC3), and with genes restricted to those above the autosomal median d_N/d_S. Confidence intervals come from 2,000 bootstrap resamples of genes within strata.

Linkage disequilibrium decays at the same rate on the Z as on the autosomes, $\rho_Z/\rho_A = 1.04$ (95% CI 0.88–1.28) from the decay fit and 1.08 (0.78–1.67) from curve rescaling, with $\rho_A = 0.026$ per base (Supplementary Table S10). Synonymous divergence is lower on the Z ($d_S = 0.298$ vs 0.358, Mann–Whitney $p = 0.009$), giving $R = 0.83$ and $\alpha = 0.33$ (95% CI 0.10–0.70), a female mutation rate about three times the male rate. Z-linked genes are slightly more constrained (median d_N/d_S 0.082 vs 0.113, $p = 0.048$) and lower in GC3 (0.36 vs 0.42, $p < 10^{-16}$). Matching on d_N/d_S leaves α unchanged (0.36), and matching on GC3 raises it to 0.43, or 0.49 when combined with d_N/d_S, so the female bias holds under each control, with the upper confidence limit reaching 1 only under joint matching. Because R estimates μ_Z/μ_A, the ratio of male Z to autosomal diversity ($\pi_Z/\pi_A = 0.00563/0.0093 = 0.61$) implies $N_{e,Z}/N_{e,A} = 0.68$–0.76 across these estimates, close to the 3/4 expected under an equal breeding sex ratio. Combined with the linkage-disequilibrium ratio, this gives $r_m/(r_m+r_f) \approx 1.02$–$1.14$, consistent with achiasmate female meiosis (Coffing et al., 2026) and inconsistent with equal recombination in the two sexes, which predicts 0.5.

Selective-sweep scan. diploS/HIC used 1.1-Mb feature windows of eleven 100-kb sub-windows and twelve diploid statistics per sub-window. It was trained on hard and soft sweeps simulated in SLiM 5.2 and recapitated with msprime, and on neutral msprime simulations, all under the fitted growth demography, at the empirical sample size, and with the empirical per-individual missing-data structure. The final model classified held-out simulations with 81% accuracy (Supplementary Fig. S7). Applied to the genome it assigns 1% of windows to the neutral class and 91% to the linked classes, because real neutral windows carry three to eight times more spatial heterogeneity in diversity among sub-windows than the simulations. We therefore used it as an outlier detector. The sweep score $p_{\text{hard}} + p_{\text{soft}}$ was ranked within strata of callable fraction (quintiles) and coding density (zero or not), and windows in the top 1% of their stratum were outliers. RAiSD and SweepFinder2 statistics were ranked the same way, and a region was concordant when a diploS/HIC outlier was supported by either. The false-discovery rate of the concordant set was estimated by shifting the joint RAiSD and SweepFinder2 outlier track circularly within each chromosome by an independent uniform offset of whole 100-kb windows, 10,000 times, and counting the diploS/HIC outlier regions that became concordant by chance (227 outlier regions, 446 footprint outlier windows among 22,278 callable windows). The mean of this null, 4.8 regions (95% range 1–9), against the 24 observed gives a false-discovery rate of 0.20 (no shift reached 24). Tier-1 calls are stratified outliers by construction and have no comparable estimate. Tier-1 calls came from a hidden Markov model with neutral, linked and sweep-centre states, whose emissions are the prior-corrected class posteriors and whose dwell lengths follow the ∼400-kb sweep footprint scaled by the recombination map. Calibrating this decoder by simulation proved infeasible, because at this population size the simulated ancestral recombination graphs carry about 3.4×10^5 marginal trees per 2.5 Mb. Its settings are therefore design-based and were checked against the concordant regions (Supplementary Fig. S13). The background-selection screen compared each call's π, from the ANGSD windows it overlaps, with the value predicted from the fraction of coding sites within 1 Mb of its midpoint, $\hat{\pi} = 0.010593 - 0.165473\, f_{\text{CDS}}$, the genome-wide linear relationship between the two. A call was retained when $\pi - \hat{\pi} \leq -0.003$, i.e. when its diversity is at least 0.003 below the background-selection expectation. Retained calls with f_{CDS} above the genome median are those well below the expectation, and those below it are low-diversity calls in gene-poor sequence where background selection is weak. Gene-ontology enrichment used genes with EnTAP (Hart et al., 2020) annotations that overlap a call, against all annotated genes outside the Z and the masked blocks.

Anomalous-diversity blocks. Blocks were called where the rolling median of 10-kb π exceeded twice the chromosome median over at least 1 Mb. Each was characterised by per-division π, between-division F_{ST}, local PCA, and genotype linkage disequilibrium, and by read depth relative to 5-Mb flanks in all 621 individuals (samtools bedcov at mapping quality 0 and 20). Long-read depth and heterozygous-site allele fractions came from PacBio HiFi reads of the assembly individual and a second individual aligned to the assembly, and genotype-likelihood π was recomputed at mapping quality 60. Blocks with population depth at least 1.15 times their flanks were classed as collapsed duplications and those at 0.90 or less as under-mapped, and the fraction of each block's accessible sequence that is not uniquely mappable was taken from GenMap (Supplementary Table S9).

References

Jeffrey R. Adrion, Jared G. Galloway, and Andrew D. Kern. Predicting the landscape of recombination using deep learning. *Molecular Biology and Evolution*, 37(6):1790–1808, 2020. doi: 10.1093/molbev/msaa038.

Nikolaos Alachiotis and Pavlos Pavlidis. RAiSD detects positive selection based on multiple signatures of a selective sweep and SNP vectors. *Communications Biology*, 1:79, 2018. doi: 10.1038/s42003-018-0085-8.

Krista D. Baker, Tony Kess, Lisa Hendrickson, Chelsea Malayny, Nicole Smith, and Ian R. Bradbury. Genomic data reveal a lack of range-wide genetic structure for a broadly distributed ommastrephid, Northern shortfin squid (*Illex illecebrosus*), in the Northwest Atlantic Ocean. *ICES Journal of Marine Science*, 82(11):fsaf201, 2025. doi: 10.1093/icesjms/fsaf201.

Tobias Baril, James Galbraith, and Alex Hayward. Earl Grey: a fully automated user-friendly transposable element annotation and analysis pipeline. *Molecular Biology and Evolution*, 41(4): msae068, 2024. doi: 10.1093/molbev/msae068.

Franz Baumdicker, Gertjan Bisschop, Daniel Goldstein, Graham Gower, Aaron P. Ragsdale, Georgia Tsambos, Sha Zhu, Bjarki Eldon, E. Castedo Ellerman, Jared G. Galloway, Ariella L. Gladstein, Gregor Gorjanc, Bing Guo, Ben Jeffery, Warren W. Kretzschmar, Konrad Lohse, Michael Matschiner, Dominic Nelson, Nathaniel S. Pope, Consuelo D. Quinto-Cortés, Murillo F. Rodrigues, Kumar Saunack, Thibaut Sellinger, Kevin Thornton, Hugo van Kemenade, Anthony W. Wohns, Yan Wong, Simon Gravel, Andrew D. Kern, Jere Koskela, Peter L. Ralph, and Jerome Kelleher. Efficient ancestry and mutation simulation with msprime 1.0. *Genetics*, 220(3):iyab229, 2022. doi: 10.1093/genetics/iyab229.

Mark A. Beaumont, Wenyang Zhang, and David J. Balding. Approximate Bayesian computation in population genetics. *Genetics*, 162(4):2025–2035, 2002. doi: 10.1093/genetics/162.4.2025.

Gaurav Bhatia, Nick Patterson, Sriram Sankararaman, and Alkes L. Price. Estimating and interpreting F_{ST}: the impact of rare variants. *Genome Research*, 23(9):1514–1521, 2013. doi: 10.1101/gr.154831.113.

Jan Boelts, Michael Deistler, Manuel Gloeckler, Álvaro Tejero-Cantero, Jan-Matthis Lueckmann, Guy Moss, Peter Steinbach, Thomas Moreau, Fabio Muratore, Julia Linhart, Conor Durkan, Julius Vetter, et al. sbi reloaded: a toolkit for simulation-based inference workflows. *Journal of Open Source Software*, 10(108):7754, 2025. doi: 10.21105/joss.07754.

Benjamin Buchfink, Klaus Reuter, and Hajk-Georg Drost. Sensitive protein alignments at tree-of-life scale using DIAMOND. *Nature Methods*, 18(4):366–368, 2021. doi: 10.1038/s41592-021-01101-x.

Christopher C. Chang, Carson C. Chow, Laurent C. A. M. Tellier, Shashaank Vattikuti, Shaun M. Purcell, and James J. Lee. Second-generation PLINK: rising to the challenge of larger and richer datasets. *GigaScience*, 4:7, 2015. doi: 10.1186/s13742-015-0047-8.

Kuan-Hao Chao, Jakob M. Heinz, Celine Hoh, Alan Mao, Alaina Shumate, Mihaela Pertea, and Steven L. Salzberg. Combining DNA and protein alignments to improve genome annotation with LiftOn. *Genome Research*, 35(2):311–325, 2025. doi: 10.1101/gr.279620.124.

Shifu Chen, Yanqing Zhou, Yaru Chen, and Jia Gu. fastp: an ultra-fast all-in-one FASTQ preprocessor. *Bioinformatics*, 34(17):i884–i890, 2018. doi: 10.1093/bioinformatics/bty560.

Gabrielle C. Coffing, Silas Tittes, Scott T. Small, Jeremea O. Songco-Casey, Denise M. Piscopo, Judit R. Pungor, Adam C. Miller, Cristopher M. Niell, and Andrew D. Kern. Cephalopod sex determination and its ancient evolutionary origin. *Current Biology*, 35(4):931–939.e4, 2025. doi: 10.1016/j.cub.2025.01.005.

Gabrielle C. Coffing, Silas Tittes, Scott T. Small, and Andrew D. Kern. Diversity and divergence of two sympatric, sibling octopus species. *Molecular Biology and Evolution*, 43(10):msag248, 2026. doi: 10.1093/molbev/msag248.

Kyle Cranmer, Johann Brehmer, and Gilles Louppe. The frontier of simulation-based inference. *Proceedings of the National Academy of Sciences*, 117(48):30055–30062, 2020. doi: 10.1073/pnas.1912789117.

Lucas Czech and Moises Exposito-Alonso. grenepipe: a flexible, scalable and reproducible pipeline to automate variant calling from sequence reads. *Bioinformatics*, 38(20):4809–4811, 2022. doi: 10.1093/bioinformatics/btac600.

Petr Danecek, James K. Bonfield, Jennifer Liddle, John Marshall, Valeriu Ohan, Martin O. Pollard, Andrew Whitwham, Thomas Keane, Shane A. McCarthy, Robert M. Davies, and Heng Li. Twelve years of SAMtools and BCFtools. *GigaScience*, 10(2):giab008, 2021. doi: 10.1093/gigascience/giab008.

Conor Durkan, Artur Bekasov, Iain Murray, and George Papamakarios. Neural spline flows. In *Advances in Neural Information Processing Systems 32 (NeurIPS 2019)*, 2019. arXiv:1906.04032.

Sean R. Eddy. Accelerated profile HMM searches. *PLoS Computational Biology*, 7(10):e1002195, 2011. doi: 10.1371/journal.pcbi.1002195.

Graham Gower, Aaron P. Ragsdale, Gertjan Bisschop, Ryan N. Gutenkunst, Matthew Hartfield, Ekaterina Noskova, Stephan Schiffels, Travis J. Struck, Jerome Kelleher, and Kevin R. Thornton. Demes: a standard format for demographic models. *Genetics*, 222(3):iyac131, 2022. doi: 10.1093/genetics/iyac131.

Benjamin C. Haller, Peter L. Ralph, and Philipp W. Messer. SLiM 5: eco-evolutionary simulations across multiple chromosomes and full genomes. *Molecular Biology and Evolution*, 43(1):msaf313, 2026. doi: 10.1093/molbev/msaf313.

Alexander J. Hart, Samuel Ginzburg, Muyang Xu, Cera R. Fisher, Nasim Rahmatpour, Jeffry B. Mitton, Robin Paul, and Jill L. Wegrzyn. EnTAP: bringing faster and smarter functional annotation to non-model eukaryotic transcriptomes. *Molecular Ecology Resources*, 20(2):591–604, 2020. doi: 10.1111/1755-0998.13106.

Julien Jouganous, Will Long, Aaron P. Ragsdale, and Simon Gravel. Inferring the joint demographic history of multiple populations: beyond the diffusion approximation. *Genetics*, 206(3):1549–1567, 2017. doi: 10.1534/genetics.117.200493.

Jerome Kelleher, Kevin R. Thornton, Jaime Ashander, and Peter L. Ralph. Efficient pedigree recording for fast population genetics simulation. *PLoS Computational Biology*, 14(11):e1006581, 2018. doi: 10.1371/journal.pcbi.1006581.

Andrew D. Kern and Daniel R. Schrider. discoal: flexible coalescent simulations with selection. *Bioinformatics*, 32(24):3839–3841, 2016. doi: 10.1093/bioinformatics/btw556.

Andrew D. Kern and Daniel R. Schrider. diploS/HIC: an updated approach to classifying selective sweeps. *G3: Genes, Genomes, Genetics*, 8(6):1959–1970, 2018. doi: 10.1534/g3.118.200262.

Thorfinn Sand Korneliussen, Anders Albrechtsen, and Rasmus Nielsen. ANGSD: analysis of next generation sequencing data. *BMC Bioinformatics*, 15:356, 2014. doi: 10.1186/s12859-014-0356-4.

Katharine L. Korunes and Mohamed A. F. Noor. Pervasive gene conversion in chromosomal inversion heterozygotes. *Molecular Ecology*, 28(6):1302–1315, 2019. doi: 10.1111/mec.14921.

Sam Kovaka, Aleksey V. Zimin, Geo M. Pertea, Roham Razaghi, Steven L. Salzberg, and Mihaela Pertea. Transcriptome assembly from long-read RNA-seq alignments with StringTie2. *Genome Biology*, 20:278, 2019. doi: 10.1186/s13059-019-1910-1.

Pablo Lemos, Adam Coogan, Yashar Hezaveh, and Laurence Perreault-Levasseur. Sampling-based accuracy testing of posterior estimators for general inference. In *Proceedings of the 40th International Conference on Machine Learning (ICML 2023)*, volume 202 of *Proceedings of Machine Learning Research*, pages 19256–19273, 2023. arXiv:2302.03026.

Heng Li. Minimap2: pairwise alignment for nucleotide sequences. *Bioinformatics*, 34(18):3094–3100, 2018. doi: 10.1093/bioinformatics/bty191.

Min Li, Baosheng Wu, Peng Zhang, Ye Li, Wenjie Xu, Kun Wang, Qiang Qiu, Jun Zhang, Jie Li, Chi Zhang, Jiangtao Fan, Chenguang Feng, and Zuozhi Chen. Genomes of two flying squid species provide novel insights into adaptations of cephalopods to pelagic life. *Genomics, Proteomics & Bioinformatics*, 20(6):1053–1065, 2022. doi: 10.1016/j.gpb.2022.09.009.

Xiaoming Liu and Yun-Xin Fu. Stairway Plot 2: demographic history inference with folded SNP frequency spectra. *Genome Biology*, 21:280, 2020. doi: 10.1186/s13059-020-02196-9.

Gilean A. T. McVean and Niall J. Cardin. Approximating the coalescent with recombination. *Philosophical Transactions of the Royal Society B: Biological Sciences*, 360(1459):1387–1393, 2005. doi: 10.1098/rstb.2005.1673.

Jiseon Min, Yuxin Ning, Nathaniel S. Pope, Franz Baumdicker, and Andrew D. Kern. Neural posterior estimation for population genetics. *Genetics*, 233(3):iyag107, 2026. doi: 10.1093/genetics/iyag107.

Jaina Mistry, Sara Chuguransky, Lowri Williams, Matloob Qureshi, Gustavo A. Salazar, Erik L. L. Sonnhammer, Silvio C. E. Tosatto, Lisanna Paladin, Shriya Raj, Lorna J. Richardson, Robert D. Finn, and Alex Bateman. Pfam: the protein families database in 2021. *Nucleic Acids Research*, 49(D1):D412–D419, 2021. doi: 10.1093/nar/gkaa913.

Takashi Miyata, Hidenori Hayashida, Kei-ichi Kuma, Kazuo Mitsuyasu, and Teruo Yasunaga. Male-driven molecular evolution: a model and nucleotide sequence analysis. *Cold Spring Harbor Symposia on Quantitative Biology*, 52:863–867, 1987. doi: 10.1101/SQB.1987.052.01.094.

Masatoshi Nei and Takashi Gojobori. Simple methods for estimating the numbers of synonymous and nonsynonymous nucleotide substitutions. *Molecular Biology and Evolution*, 3(5):418–426, 1986. doi: 10.1093/oxfordjournals.molbev.a040410.

Tomoko Ohta and Motoo Kimura. Linkage disequilibrium between two segregating nucleotide sites under the steady flux of mutations in a finite population. *Genetics*, 68(4):571–580, 1971. doi: 10.1093/genetics/68.4.571.

Francisco J. Pardo-Palacios, Angeles Arzalluz-Luque, Liudmyla Kondratova, Pedro Salguero, Jorge Mestre-Tomás, Rocío Amorín, Eva Estevan-Morió, Tianyuan Liu, Adalena Nanni, Lauren McIntyre, Elizabeth Tseng, and Ana Conesa. SQANTI3: curation of long-read transcriptomes for accurate identification of known and novel isoforms. *Nature Methods*, 21(5):793–797, 2024. doi: 10.1038/s41592-024-02229-2.

Stephan Peischl, Eric Koch, Rafael F. Guerrero, and Mark Kirkpatrick. A sequential coalescent algorithm for chromosomal inversions. *Heredity*, 111(3):200–209, 2013. doi: 10.1038/hdy.2013.38.

Christopher Pockrandt, Mai Alzamel, Costas S. Iliopoulos, and Knut Reinert. GenMap: ultra-fast computation of genome mappability. *Bioinformatics*, 36(12):3687–3692, 2020. doi: 10.1093/bioinformatics/btaa222.

Nathaniel S. Pope, Angel G. Rivera-Colón, Ananya Kapoor, Kevin Korfmann, Murillo F. Rodrigues, Scott T. Small, Anastasia A. Teterina, and Andrew D. Kern. GPU accelerated population genetics statistics using pg_gpu. *bioRxiv*, 2026. doi: 10.64898/2026.05.29.728868. Preprint.

Ryan Poplin, Valentin Ruano-Rubio, Mark A. DePristo, Tim J. Fennell, Mauricio O. Carneiro, Geraldine A. Van der Auwera, David E. Kling, Laura D. Gauthier, Ami Levy-Moonshine, David Roazen, Khalid Shakir, Joel Thibault, Sheila Chandran, Chris Whelan, Monkol Lek, Stacey Gabriel, Mark J. Daly, Ben Neale, Daniel G. MacArthur, and Eric Banks. Scaling accurate genetic variant discovery to tens of thousands of samples. *bioRxiv*, page 201178, 2017. doi: 10.1101/201178. Preprint.

Pierre Pudlo, Jean-Michel Marin, Arnaud Estoup, Jean-Marie Cornuet, Mathieu Gautier, and Christian P. Robert. Reliable ABC model choice via random forests. *Bioinformatics*, 32(6):859–866, 2016. doi: 10.1093/bioinformatics/btv684.

Aaron R. Quinlan and Ira M. Hall. BEDTools: a flexible suite of utilities for comparing genomic features. *Bioinformatics*, 26(6):841–842, 2010. doi: 10.1093/bioinformatics/btq033.

Aaron P. Ragsdale and Simon Gravel. Unbiased estimation of linkage disequilibrium from unphased data. *Molecular Biology and Evolution*, 37(3):923–932, 2020. doi: 10.1093/molbev/msz265.

Vincent Ranwez, Emmanuel J. P. Douzery, Cédric Cambon, Nathalie Chantret, and Frédéric Delsuc. MACSE v2: toolkit for the alignment of coding sequences accounting for frameshifts and stop codons. *Molecular Biology and Evolution*, 35(10):2582–2584, 2018. doi: 10.1093/molbev/msy159.

Angel G. Rivera-Colón, Clara T. Rehmann, and Andrew D. Kern. MKado: a toolkit for McDonald–Kreitman tests of natural selection. *G3: Genes, Genomes, Genetics*, 16(8):jkag142, 2026. doi: 10.1093/g3journal/jkag142.

Enrique Santiago, Carlos Köpke, and Armando Caballero. Accounting for population structure and data quality in demographic inference with linkage disequilibrium methods. *Nature Communications*, 16:6054, 2025. doi: 10.1038/s41467-025-61378-w.

Scott T. Small. msinv: an ARG-based coalescent simulator with chromosomal inversions. https://github.com/stsmall/msinv, 2025. Software, version 0.4.1.

Scott T. Small, Silas Tittes, Thomas Desvignes, John H. Postlethwait, and Andrew D. Kern. Dosage compensation of the Z chromosome is ancestral and conserved across coleoid cephalopods. Manuscript in preparation, 2026.

Baoxing Song, Santiago Marco-Sola, Miquel Moreto, Lynn Johnson, Edward S. Buckler, and Michelle C. Stitzer. AnchorWave: sensitive alignment of genomes with high sequence diversity, extensive structural polymorphism, and whole-genome duplication. *Proceedings of the National Academy of Sciences*, 119(1):e2113075119, 2022. doi: 10.1073/pnas.2113075119.

Sean Talts, Michael Betancourt, Daniel Simpson, Aki Vehtari, and Andrew Gelman. Validating Bayesian inference algorithms with simulation-based calibration, 2018. arXiv:1804.06788.

Alvaro Tejero-Cantero, Jan Boelts, Michael Deistler, Jan-Matthis Lueckmann, Conor Durkan, Pedro J. Gonçalves, David S. Greenberg, and Jakob H. Macke. sbi: A toolkit for simulation-based inference. *Journal of Open Source Software*, 5(52):2505, 2020. doi: 10.21105/joss.02505.

Gregg W. C. Thomas. degenotate: annotation of degeneracy of coding sites. https://github.com/harvardinformatics/degenotate, 2024. Software, version 1.3.

Md. Vasimuddin, Sanchit Misra, Heng Li, and Srinivas Aluru. Efficient architecture-aware acceleration of BWA-MEM for multicore systems. In *2019 IEEE International Parallel and Distributed Processing Symposium (IPDPS)*, pages 314–324, 2019. doi: 10.1109/IPDPS.2019.00041.

Supplementary figures

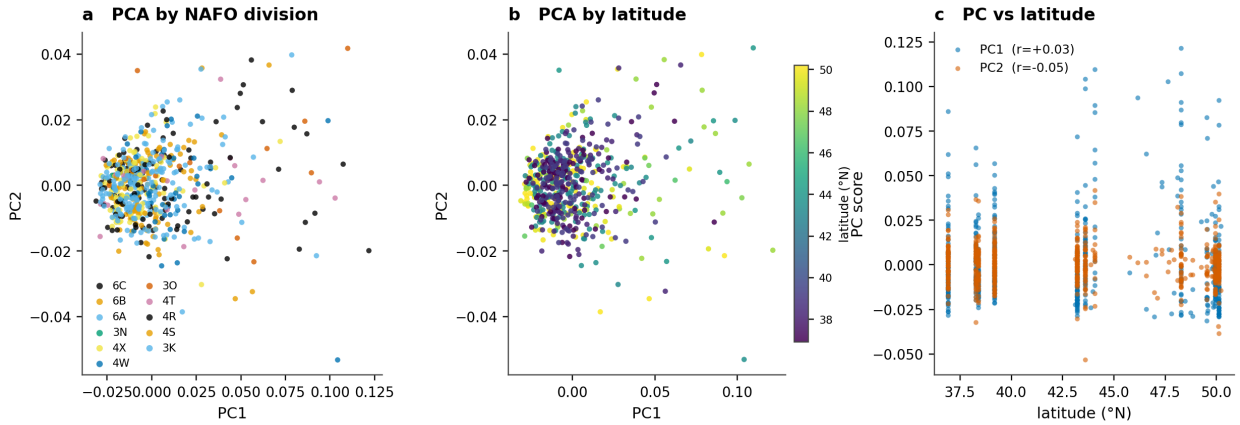


Figure S1: **The autosomal-PCA spread is not geographic structure.** Genome-wide PCA (all autosomes except chromosome 2, 280,131 LD-thinned SNPs), $n = 619$ individuals. (a) Coloured by NAFO sampling division and (b) by latitude, the divisions and the latitudinal gradient are fully intermixed across PC space, with no clustering by origin. (c) PC scores plotted against latitude show no relationship (Pearson $r = 0.03$ for PC1, $r = -0.05$ for PC2). PC1 explains 1.4% of the variance and tracks sequencing depth (Spearman $\rho = -0.99$ with the number of autosomal reads). PC2 explains 0.4% and is carried by four individuals (80% of its variance).

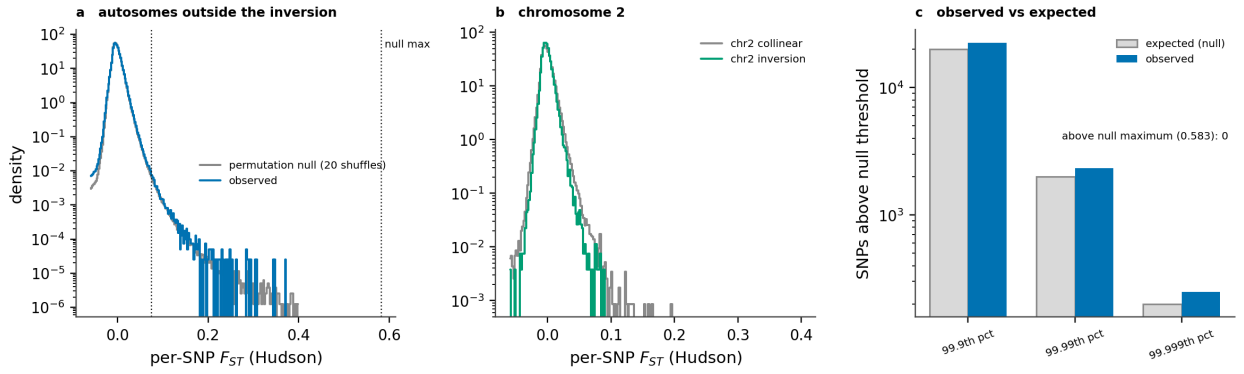


Figure S2: **No geographic differentiation beyond genetic drift and no local adaptation signal.** Per-SNP Hudson F_{ST} across all 45 NAFO-division pairs for every biallelic SNP with minor-allele frequency above 0.05 ($\geq$40 genotyped individuals per pair, ratio of averages over $\geq$20 informative pairs). **(a)** On the autosomes outside the inversion (19,984,027 SNPs), the empirical distribution (blue) matches the permutation null (grey), built by shuffling division labels among individuals 20 times, which keeps genotype calling, missingness, the allele-frequency spectrum and linkage. Empirical maximum $F_{\mathrm{ST}} = 0.371$ against a null maximum of 0.583. **(b)** On chromosome 2, the inverted region (green) is no more geographically differentiated than the collinear remainder, so the inversion is non-clinal and its $F_{\mathrm{ST}} \approx 0.35$ is between karyotypes, not geography. Its upper tail is lighter than the null (66 SNPs above the null 99.9th percentile against 134 expected). **(c)** SNPs above the null 99.9th, 99.99th and 99.999th percentiles (22,562, 2,319 and 249) are within the permutation range ($19{,}979 \pm 2{,}154$, $1{,}998 \pm 352$ and 200 ± 54, permutation $p = 0.19$, 0.24 and 0.29), and no SNP exceeds the null maximum, so a scan for local adaptation that the diversity-based sweep tests could miss returns no candidates. Because the minor-allele-frequency filter is applied to the pooled 620 individuals, a variant private to one small division can fall below it (a SNP fixed only in 3O, with 16 individuals, has a pooled frequency of 2.5%), so the scan rules out strongly differentiated SNPs that are common overall, but not private alleles in the smallest divisions.

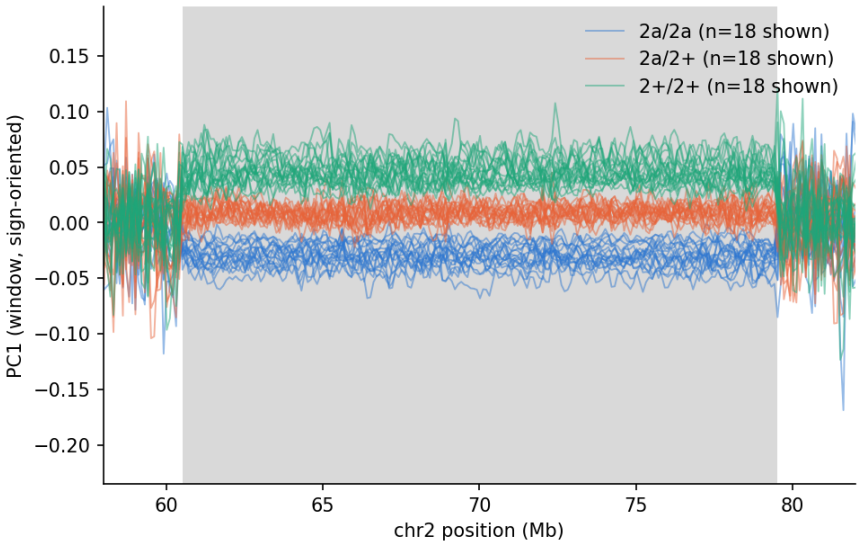


Figure S3: **Windowed PCA across chromosome 2** localising the differentiated block.

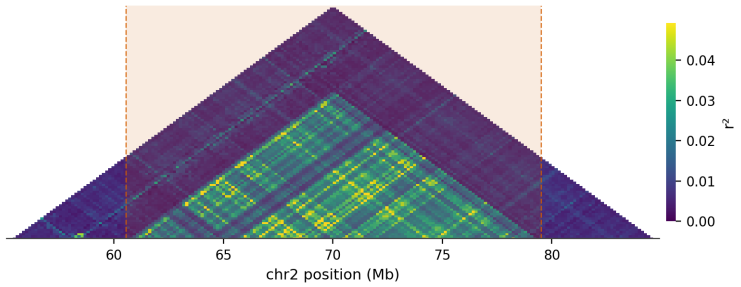


Figure S4: **Linkage disequilibrium across the chromosome-2 inversion.** Mean genotype r^2 between SNPs, binned along chr2:55–85 Mb, pooling all 621 individuals. Dashed lines and shading mark the breakpoints (60.54 and 79.50 Mb). A block of elevated linkage disequilibrium spans exactly the inversion and stops at both breakpoints, while the flanking collinear sequence shows only background values. Mean r^2 between SNPs more than 1 Mb apart is 3.5 times higher inside the inversion than in its flanks (0.0056 vs 0.0016, Supplementary Fig. S24). This long-range association arises because the few near-fixed differences between arrangements are all correlated with karyotype, and it is the direct signature of suppressed recombination between arrangements. Recombination within arrangements is not suppressed at the scale of the 3–6-kb windows used by the recombination map, which is why the inversion is shown here rather than in Supplementary Fig. S12.

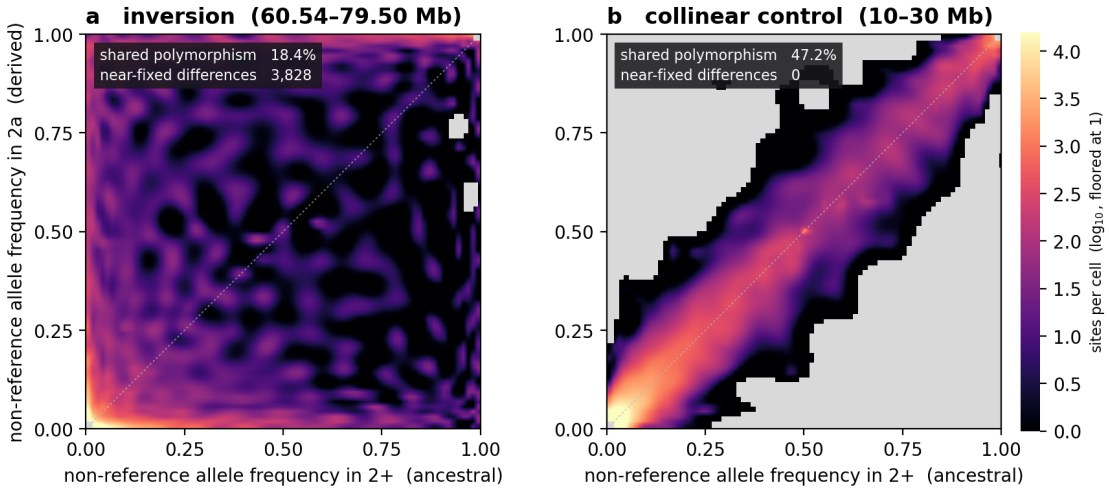


Figure S5: **Joint two-dimensional site-frequency spectrum** of non-reference allele frequency in the 2a (derived) versus 2+ (ancestral) arrangement. Each cell is a count of sites (colour, $\log_{10}$). In the inversion (a) density spreads off the diagonal and into the corners, with variants common in one arrangement and absent in the other, i.e. near-fixed differences (3,828 sites, 18% shared polymorphism), the direct signature of the recombination barrier. In the collinear control (b) the density collapses onto the diagonal (0 near-fixed differences, 47% shared), as expected for freely recombining, exchangeable sequence.

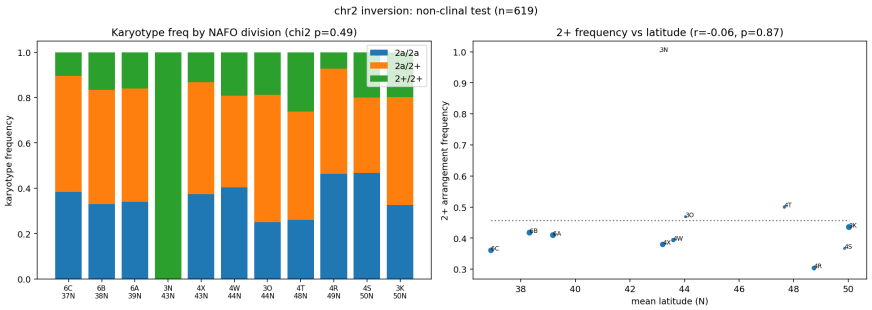


Figure S6: **The inversion is non-clinal.** Frequency of the ancestral 2+ arrangement across NAFO sampling divisions shows no latitudinal trend (see table S2). Geographic $F_{ST} \approx 0$, and the three karyotypes co-occur throughout the range.

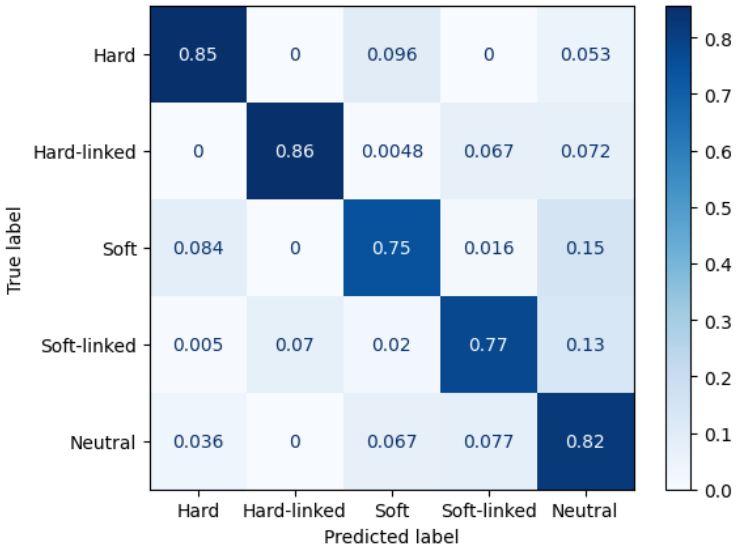


Figure S7: **Sweep-classifier confusion matrix** (five-class diploSHIC, full-SFS model, 81% overall accuracy). Recall is 0.85 for hard, 0.86 for linked-hard, 0.82 for neutral, 0.75 for soft and 0.77 for linked-soft sweeps.

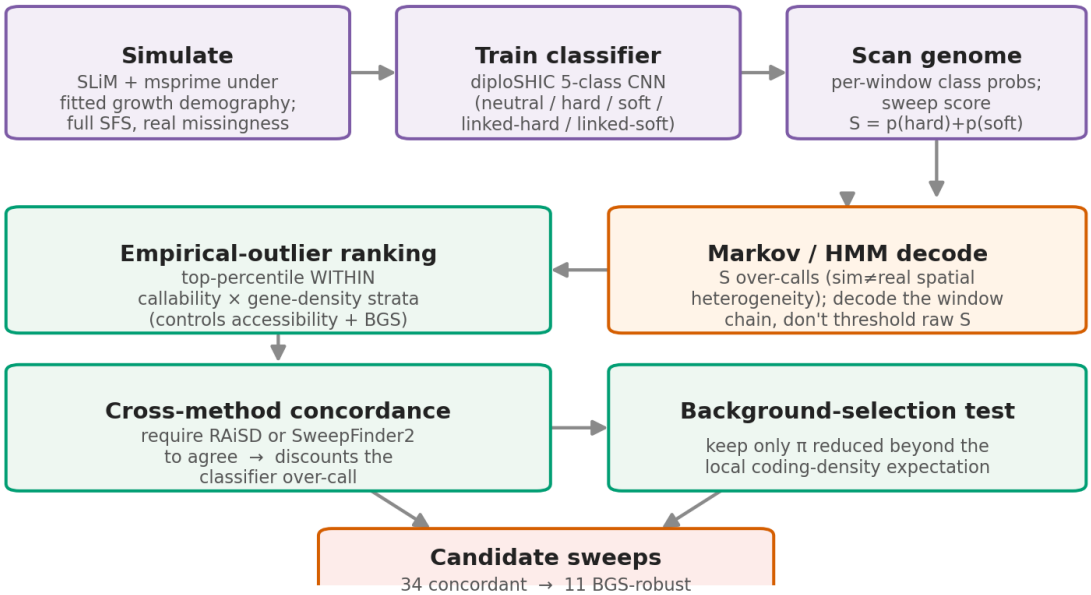


Figure S8: **The selective-sweep scan pipeline.** Simulate (SLiM + msprime under the fitted growth demography) → diploSHIC five-class classifier → Markov/HMM decode → stratified empirical-outlier ranking (within callability × gene-density strata) + cross-method concordance (RAiSD, SweepFinder2) + background-selection filter → candidate sweeps.

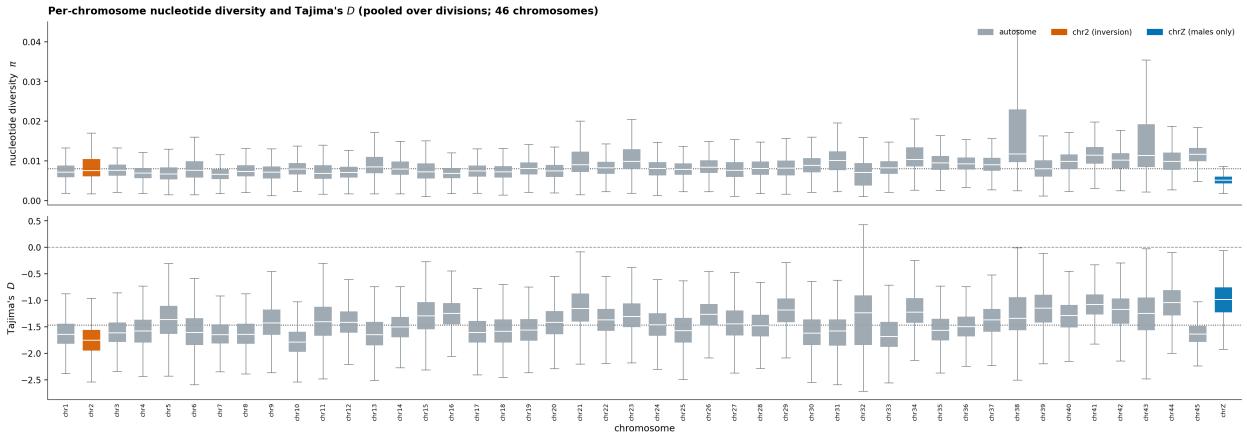


Figure S9: **Per-chromosome diversity.** Distribution of nucleotide diversity π (top) and Tajima's D (bottom) across 50 kb windows, one box per chromosome (pooled over the ten NAFO divisions, for all 46 chromosomes, the 45 autosomes plus chrZ). Diversity is uniform and D is strongly negative genome-wide, the signature of expansion. Two chromosomes stand apart. Chromosome 2 (the inversion, orange) carries an inflated upper tail of π and more-negative D from the segregating arrangements, and chrZ (blue, computed in the 329 hemizygous-informative males only, as females are Z-haploid) has reduced diversity (π median 0.005 vs ∼0.008 autosomal, ≈0.63×) as expected for a sex chromosome.

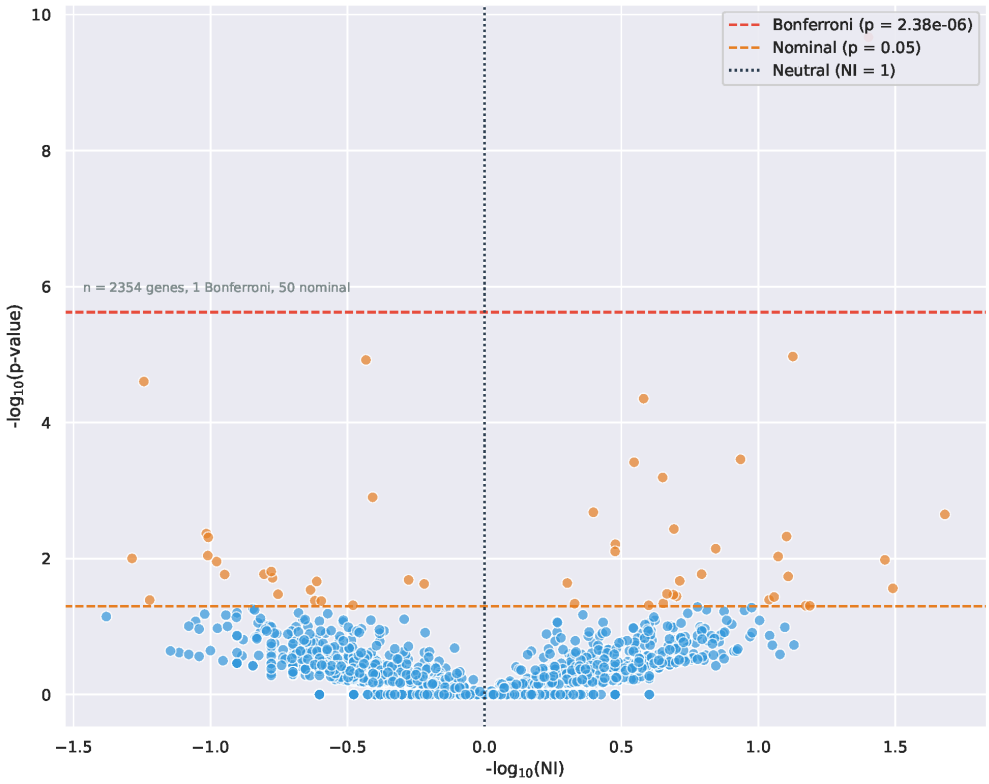


Figure S10: **Per-gene McDonald–Kreitman neutrality index** (mkado volcano plot, *I. coindetii* outgroup). Each point is one gene, plotted as $-\log_{10}$ of its neutrality index $\mathrm{NI} = (P_n/P_s)/(D_n/D_s)$ on the x-axis (right of the neutral $\mathrm{NI} = 1$ line = adaptive, $\mathrm{NI} < 1$, left = excess non-synonymous polymorphism, $\mathrm{NI} > 1$) against $-\log_{10} p$ on the y-axis. Dashed lines mark the nominal ($p = 0.05$) and Bonferroni thresholds. Of the 20,999 genes in the test, the 2,354 with non-zero D_n, D_s, P_n and P_s have a finite NI and are plotted. Of these, 51 are nominally significant, and one of them also passes Bonferroni correction over all 20,999 genes ($p < 2.4 \times 10^{-6}$). The genome-wide asymptotic $\alpha = 0.64$ resolves to few individual genes, as expected for a large-N_e species where per-gene MK is underpowered.

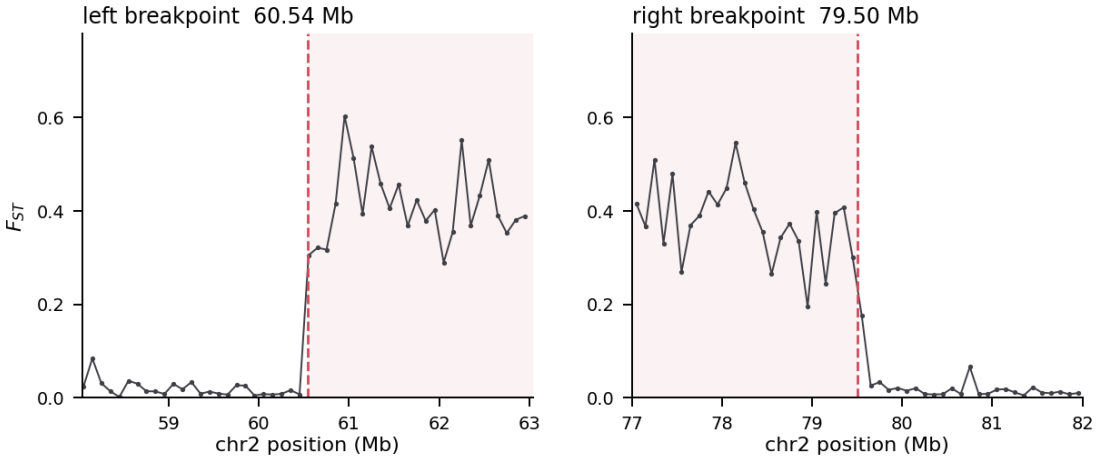


Figure S11: **Inversion breakpoints.** F_{ST} across the left (60.54 Mb) and right (79.50 Mb) breakpoints, a step rather than a ramp.

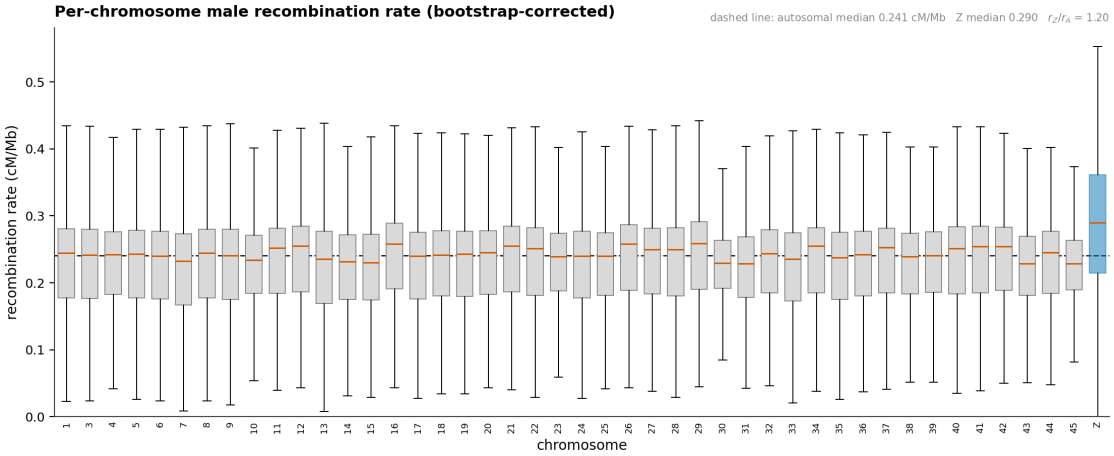


Figure S12: **Per-chromosome male recombination-rate distribution** (ReLERNN). Rates are corrected with 200 parametric-bootstrap replicates in each of 20 rate slices. The autosomes were predicted in 3–6-kb windows by one network, trained on chromosomes 1, 13 and 45 and applied to every autosome except chromosome 2. The Z was predicted in 30-kb windows by a separate network trained with the Z effective size set to 0.60 of the autosomal value (the Z-to-autosome diversity ratio), which places its rate on the autosomal scale. The dashed line marks the median of all autosomal windows (0.241 cM/Mb), and per-chromosome medians span 0.229–0.259 cM/Mb. The Z median is 0.290 cM/Mb, so $r_Z/r_A = 1.20$. Because the Z recombines only in ZZ males, equal recombination in the two sexes predicts $r_Z/r_A = 0.67$ and achiasmate female meiosis predicts 1.33. Chromosome 2 is not shown, because a window-scale map estimated from all karyotypes together cannot resolve a barrier that acts only between arrangements. That barrier is shown directly by the linkage-disequilibrium block spanning the inversion (Supplementary Fig. S4).

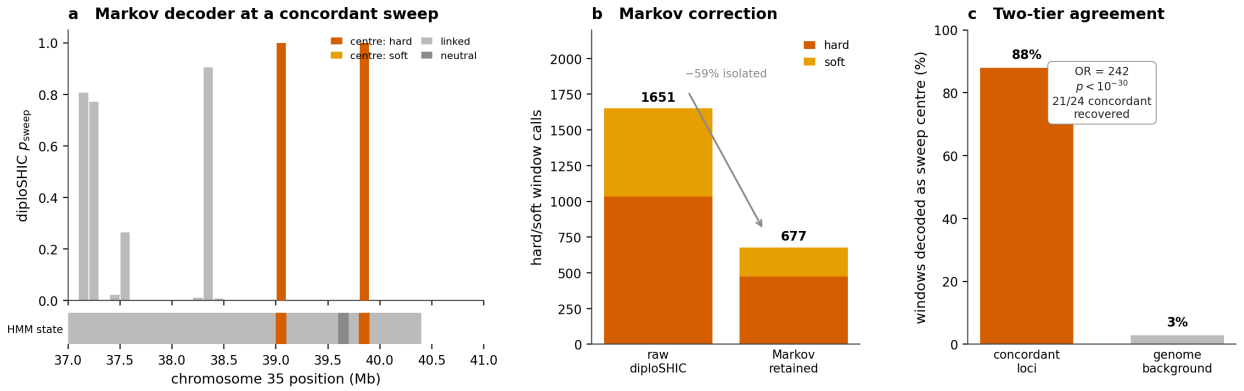


Figure S13: **Recombination-aware Markov decoder for the Tier-1 sweep calls.** A hidden Markov model over the per-100 kb-window diploSHIC classification (emissions = prior-corrected class posteriors, dwell lengths from the ∼400 kb sweep footprint scaled by the ReLERNN map, design-based rather than simulation-calibrated). **(a)** The decoder at a representative concordant sweep (chromosome 35, *ZEB2*), showing the sweep posterior $p_{\text{hard}} + p_{\text{soft}}$ per window and the Viterbi state track (neutral / linked / centre). It localises the centre inside the linked envelope and overrides spatially isolated high-posterior windows (e.g. near 38.35 Mb). **(b)** Markov correction reduces the raw hard/soft windows from 1,651 to 677 spatially coherent windows (−59%), which merge into 623 localized calls. **(c)** 88% of windows at the Tier-2 cross-method-concordant loci are decoded as sweep centres versus 2.9% genome-wide (odds ratio 242, $p < 10^{-30}$, 21 of 24 concordant regions recovered).

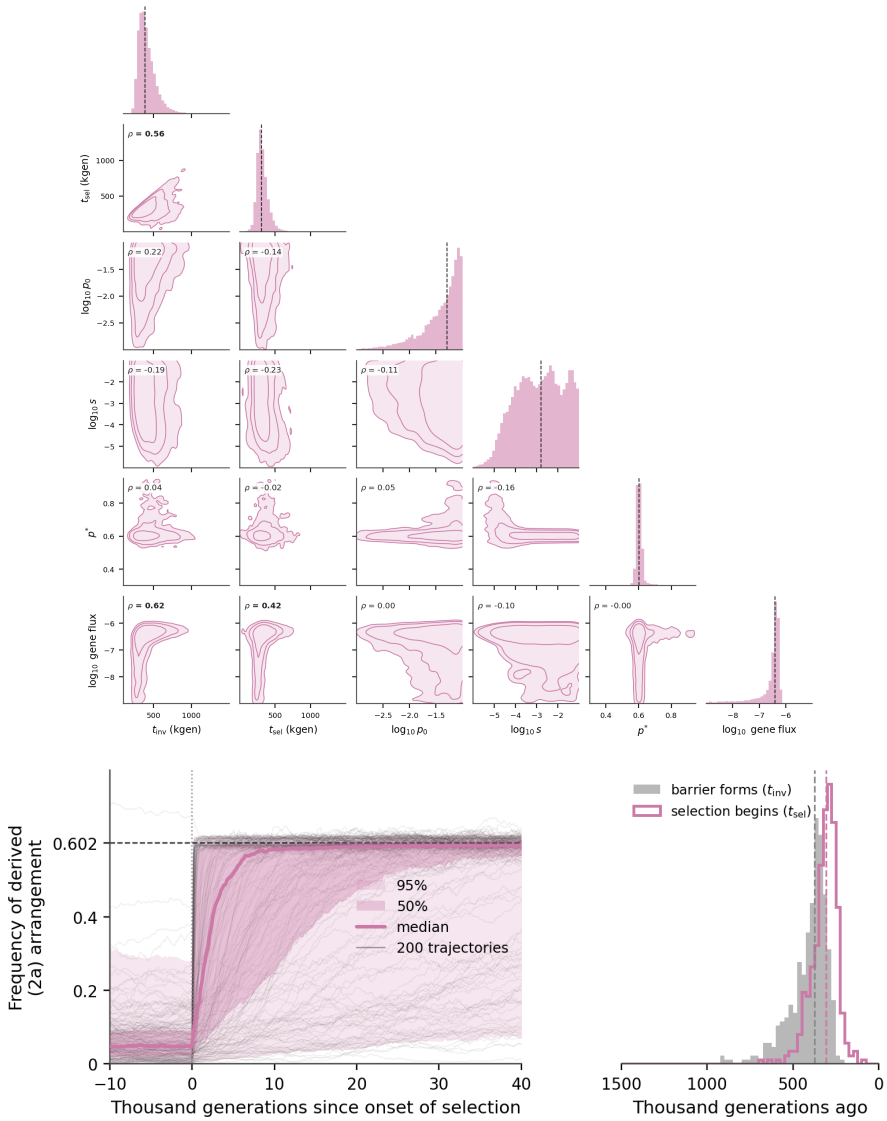


Figure S14: **Posterior of the balancing model with gene flux for the chromosome-2 inversion** (main-text Fig. 4b). Top, joint posterior from neural posterior estimation, with each axis spanning its prior. The diagonal gives the marginals with medians dashed, and below it the contours enclose 80, 95 and 99% of the posterior, with the Spearman correlation in each panel (bold where $|\rho| \geq 0.3$). The barrier age t_{inv} and the onset of selection t_{sel} are in thousands of generations. All draws are soft starts, which carry 97% of the posterior mass of balancing selection. The selection coefficient has only a lower bound and follows the prior above about 10^{-4}. The gene-flux rate is the per-site conversion probability per generation in heterokaryotypes. Bottom left, frequency of 2a in 200 trajectories re-simulated from posterior draws, aligned on the onset of selection, with the pointwise median, 50% and 95% ranges of 392. The median trajectory reaches 0.55 about 6,000 generations after onset, and the slowest fifth, mostly those with $s < 10^{-4}$, take more than 50,000 generations. Bottom right, the posterior draws of t_{inv} and t_{sel} behind these trajectories, with medians dashed.

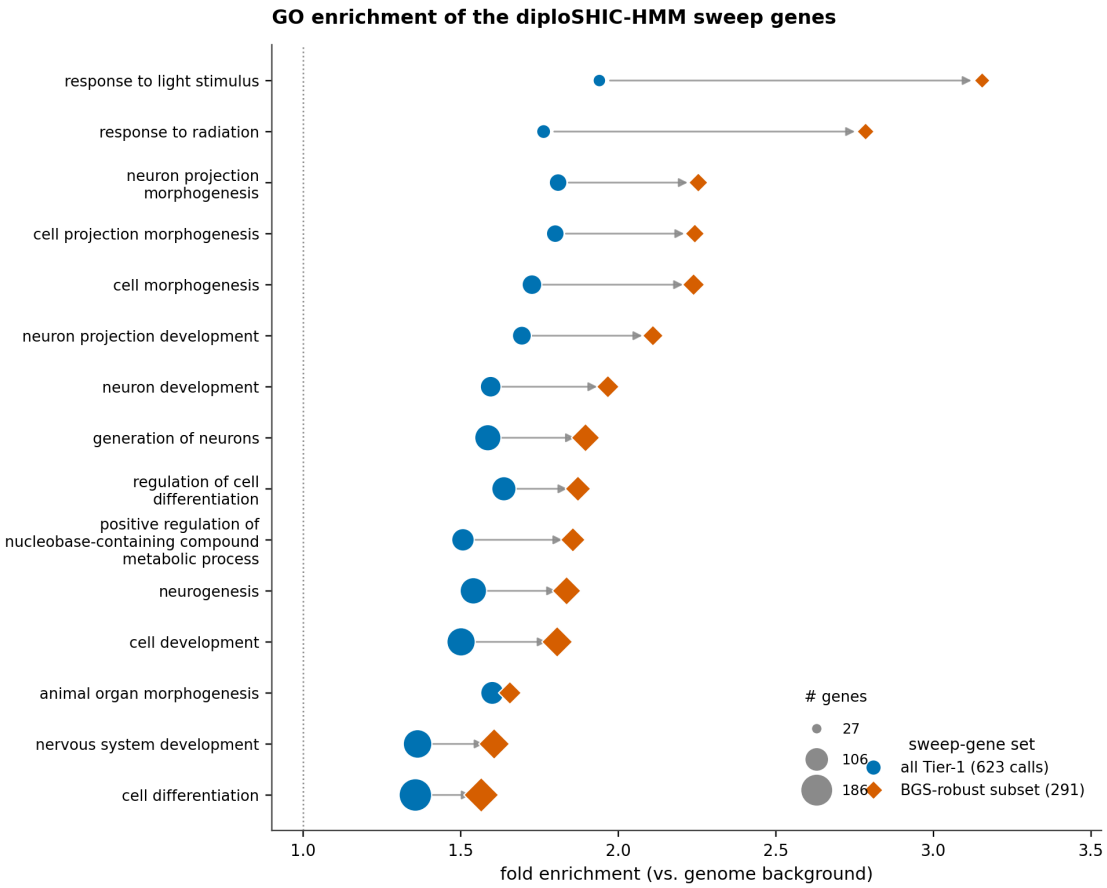


Figure S15: **GO enrichment of the diploSHIC-HMM sweep genes** (biological process, terms significant at FDR<0.05 in both sets). In this paired dot plot each term is shown for the full Tier-1 set (blue circle, 623 calls) and for its background-selection-robust subset (orange diamond, 291 calls), with x = fold enrichment and marker area = number of sweep genes carrying the term. The grey arrow marks the shift between the two sets. The BGS screen consistently sharpens the enrichment (rightward), and most strongly for the light-response / photoreception terms (response to light stimulus 1.9 $\rightarrow$ 3.2×, response to radiation 1.8 $\rightarrow$ 2.8×). Two coherent themes dominate, neural development and photoreception, and both survive the background-selection filter (table S4, full per-set enrichment tables are provided with the source). The hard-sweep calls are a sub-classification recorded in the calls table, not a separate gene set.

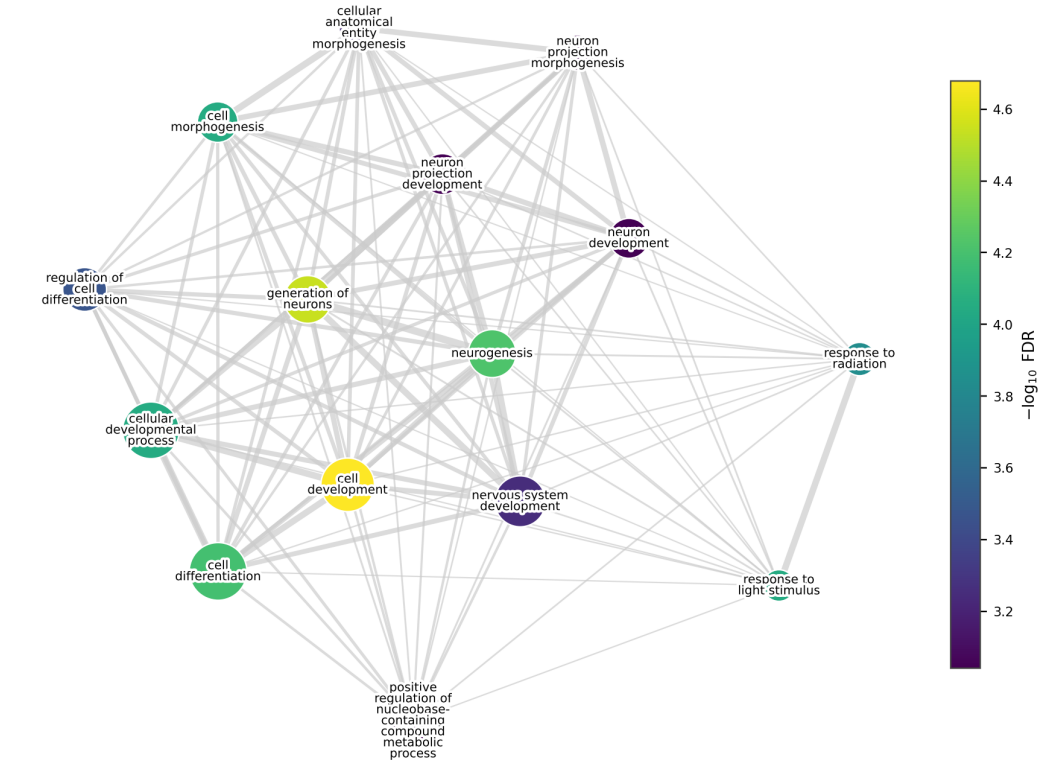


Figure S16: **GO enrichment map for the background-selection-robust sweep genes.** Built from the 291 BGS-robust Tier-1 calls (the subset whose local diversity reduction exceeds the background-selection expectation, 227 GO-annotated genes). Nodes are enriched biological-process terms (size = number of genes, colour = $-\log_{10}$ FDR). Edges connect terms sharing genes (Jaccard ≥ 0.15). The terms form connected modules of neural development / cell morphogenesis and photoreception (response to light/radiation), the same two themes seen in the GSEA dot plot (fig. S15).

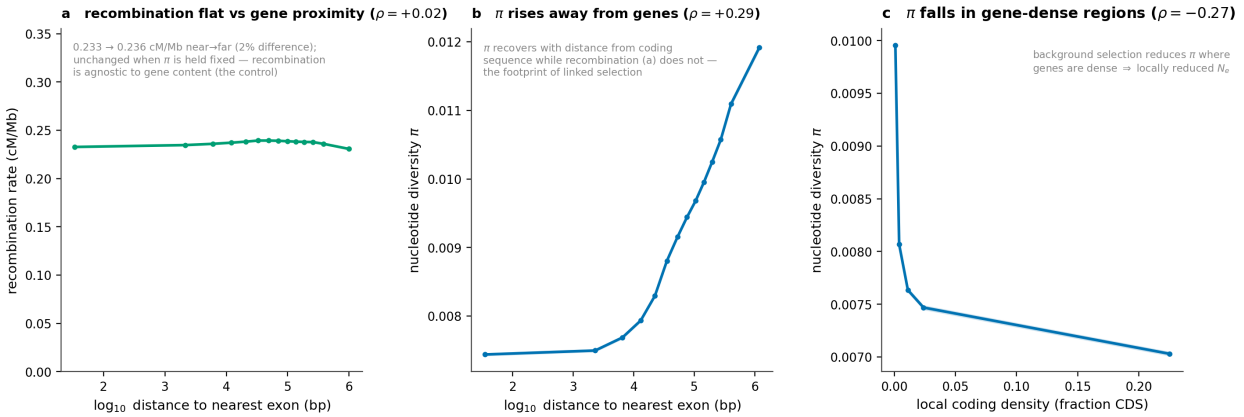


Figure S17: **Recombination, gene proximity and the efficacy of selection.** Per-window statistics (∼50 kb windows, autosomes). **(a)** As a control, recombination rate is essentially flat across distance from the nearest exon (0.233→0.236 cM/Mb near→far, a ∼2% difference, Spearman $\rho = +0.02$, unchanged when π is held fixed, partial $\rho = -0.01$), so recombination is agnostic to gene content, as it should be. **(b)** Nucleotide diversity π, by contrast, rises steeply with distance from coding sequence ($\rho = +0.29$, +43% near→far), the footprint of linked selection. **(c)** π falls with local coding density ($\rho = -0.27$), as background selection reduces diversity where genes are dense, i.e. a locally reduced effective size, and it is gene-density-driven. Linked selection in *Illex illecebrosus* is thus structured by gene density and gene proximity rather than by the recombination-rate axis, which is nearly uniform genome-wide (per-chromosome medians 0.23–0.26 cM/Mb, Supplementary Figs. S12, S22).

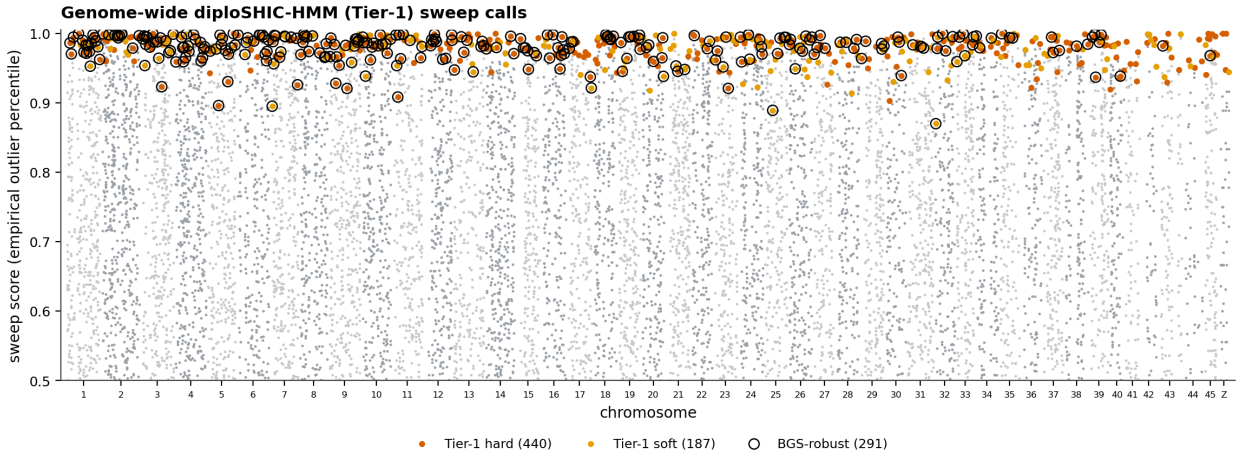


Figure S18: **Genome-wide diploSHIC-HMM (Tier-1) sweep calls.** Per-100 kb-window empirical-outlier sweep score (rank within callability × coding-density strata) across all 45 autosomes plus the Z chromosome (627 calls in total, 440 hard and 187 soft). The Markov-decoded Tier-1 calls are highlighted by class (hard, soft), and the background-selection-robust subset (§ BGS screen, 291 of the 623 autosomal calls) is ringed. The four chromosome-Z calls come from the male-only (n=329) diploSHIC scan and are diploSHIC-only (RAiSD/SweepFinder2 do not cover the Z). The calls are distributed across the genome with no single dominant locus.

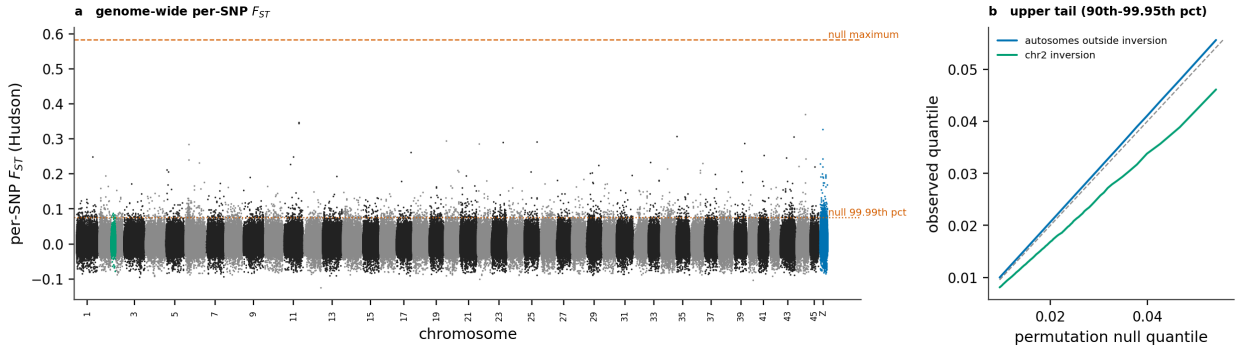


Figure S19: **Genome-wide per-SNP F_{ST} finds no differentiation outlier.** Per-SNP Hudson F_{ST} of fig. S2 along the genome (20,276,930 SNPs with minor-allele frequency above 0.05 that are informative in at least 20 division pairs, 20,117,984 on the autosomes and 158,946 on the Z, every SNP shown). **(a)** No SNP exceeds the permutation-null maximum (0.583, dashed). The 2,327 SNPs above the null 99.99th percentile (0.075, dotted) on the autosomes match the number expected by chance and are spread across all chromosomes. The chromosome-2 inversion is green. chrZ (blue) exceeds the null tail (187 SNPs above the 99.99th percentile against 16 expected) because division sex ratios differ and females are hemizygous, and it is shown for completeness. **(b)** Upper-tail quantiles of the observed against the permutation null. The autosomes outside the inversion follow the identity line, and the inversion lies below it.

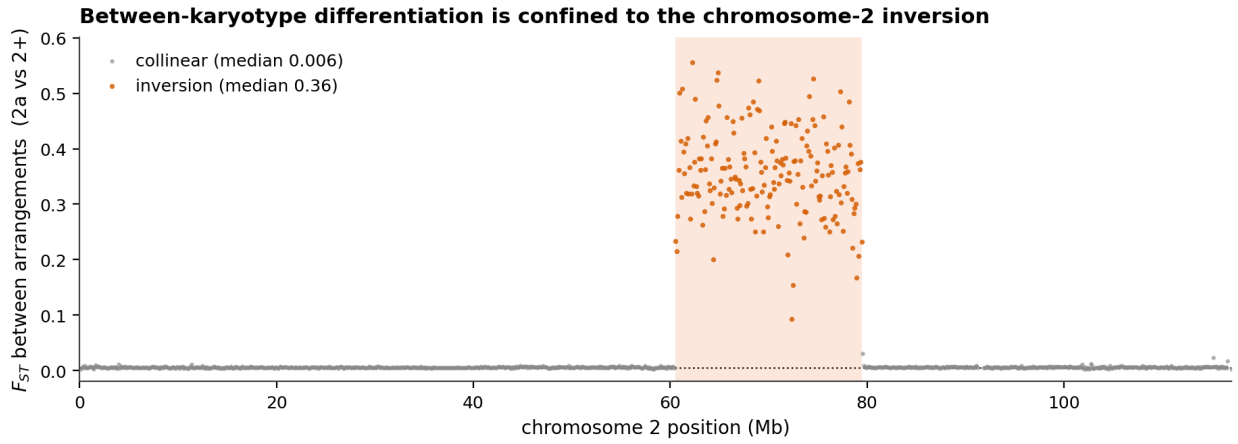


Figure S20: **Between-karyotype differentiation is confined to the chromosome-2 inversion.** Per-window F_{ST} between the 2a and 2+ arrangements (arrangement homozygotes, 2a/2a $n = 223$ and 2+/2+ $n = 97$) across the whole of chromosome 2, aggregated in 100 kb windows (ratio-of-averages Hudson estimator). Unlike the population-F_{ST} scan (which asks about geography), this asks how differentiated the two arrangements are along the chromosome. The collinear remainder (grey) sits at $F_{ST} \approx 0$ (median 0.006), so outside the inversion the karyotypes are one panmictic population, while the inversion (60.54–79.50 Mb, shaded) is strongly differentiated (median 0.36, max 0.56). The whole-inversion F_{ST} of 0.35 that defines the "structure" on chromosome 2 is thus between karyotypes and confined to the inversion, not geographic.

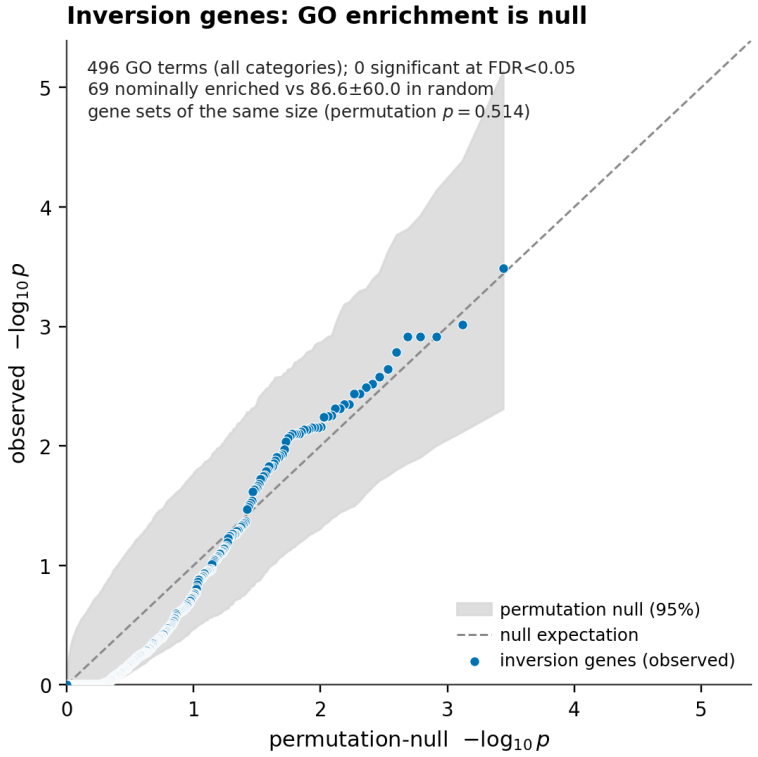


Figure S21: **The inversion-captured genes show no GO enrichment.** QQ-plot of the per-term GO-enrichment p-values (Fisher's exact vs. genome background, all three ontologies) for the genes in the inversion against a permutation null (500 random gene sets of the same size). A theoretical-uniform null is inappropriate for GO, because nested and correlated terms and tiny discrete counts inflate any small gene set, so the permutation band is the correct reference. The observed points fall entirely within it, and the inversion genes yield fewer nominally enriched terms than random sets of the same size (69 vs 86.6 ± 60, permutation $p = 0.51$) and none survive FDR<0.05. The large per-term fold values (up to $\sim$40$\times$) are small-count artefacts (most rest on 2 of the 29 GO-annotated genes), not enrichment (table S7).

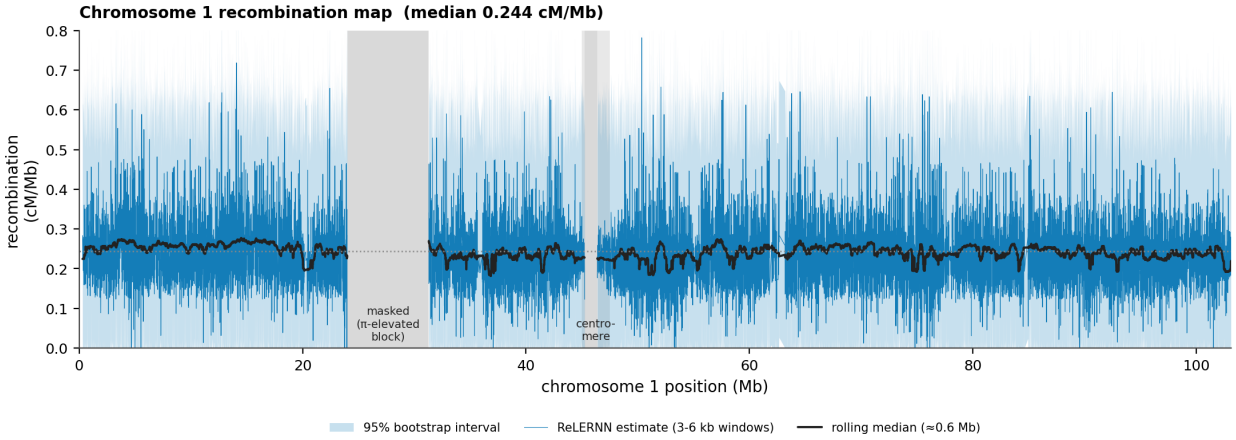


Figure S22: **The chromosome-1 recombination map, and what the LD-based map does not resolve.** Male ReLERNN estimate in 3–6-kb windows, corrected with 200 bootstrap replicates, with 95% bootstrap intervals and a 0.6-Mb rolling median (median 0.244 cM/Mb). Two stretches carry no windows. The π-elevated block at 24.0–31.3 Mb was masked. The 1.1-Mb gap at 45.25–46.39 Mb lies inside the 2.5-Mb block at 45.0–47.5 Mb, which is 80% repeat sequence (vs 52% chromosome-wide) and is the centromere. In the accessible pericentromeric flanks the estimate dips only $\sim$7%, although nucleotide diversity halves there (0.0038 vs 0.0084), and across the whole 103 Mb the 2-Mb rolling median spans just 0.22–0.27 cM/Mb ($-$10% to +10% of the median). The map's chromosome-scale dynamic range is therefore compressed, so centromeric suppression cannot be resolved and the nearly uniform landscape is in part an estimator ceiling.

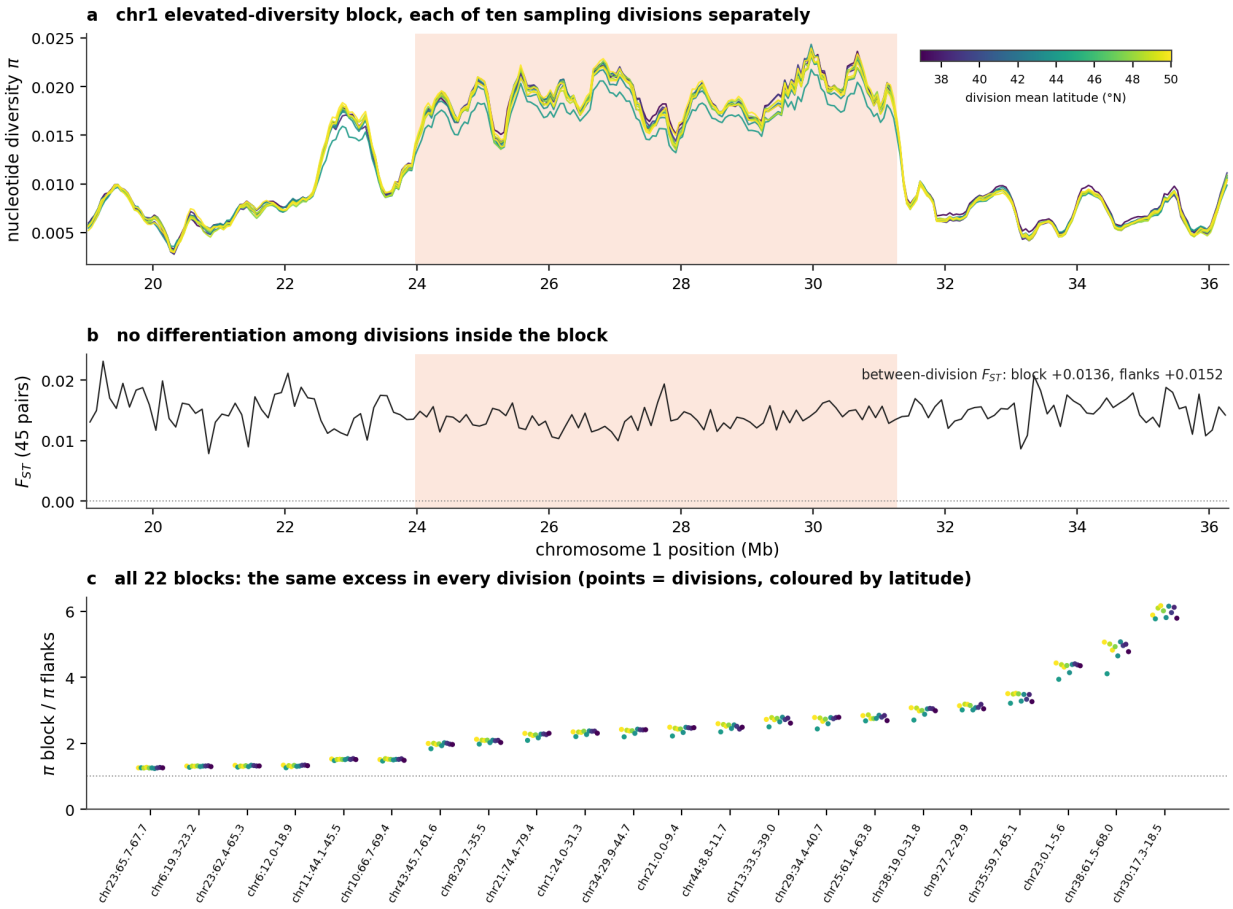


Figure S23: **The 22 elevated-diversity blocks are shared by every sampling division.** **(a)** Nucleotide diversity across chr1:19–36 Mb computed separately in each of the ten divisions with more than one individual (ANGSD, 50-kb windows, lines coloured by division latitude). The 24.0–31.3 Mb block (shaded) carries 2.2–2.4 times the flanking diversity in every division. **(b)** Between-division Hudson F_{ST} over the same span (all 45 division pairs, 100-kb windows) is lower inside the block than in the flanks (0.014 vs 0.015). The same holds for the three strongest blocks (chr30:17.3–18.5, chr23:0.1–5.6 and chr38:61.5–68.0 Mb, 0.006–0.007 inside vs 0.014–0.015 outside). **(c)** Block-to-flank diversity ratio per division for all 22 blocks. The excess varies by only 1–6% among divisions, so no block is driven by a subset of the sample. Read depth separates the blocks into collapsed duplications (depth raised in every individual, up to 2.2 times the flanks in the population sample and 2.7 and 2.4 times in long reads from the assembly individual and a second individual, with five to six times the flanking density of heterozygous sites at skewed allele fractions in single diploid genomes), under-mapped blocks (short-read depth 0.6–0.9 times the flanks, a third of the accessible sequence not uniquely mappable, and a much smaller deficit in long reads), and gene-poor regions of elevated diversity with normal depth (Supplementary Table S9). Genotype-likelihood diversity at mapping quality 60 retains most of the excess (for example chr30 6.4 to 5.8 times, chr23 5.2 to 4.0 times), because reads from unassembled copies map uniquely to the single assembled copy. All 22 blocks are masked from the sweep scan and gene-set analyses.

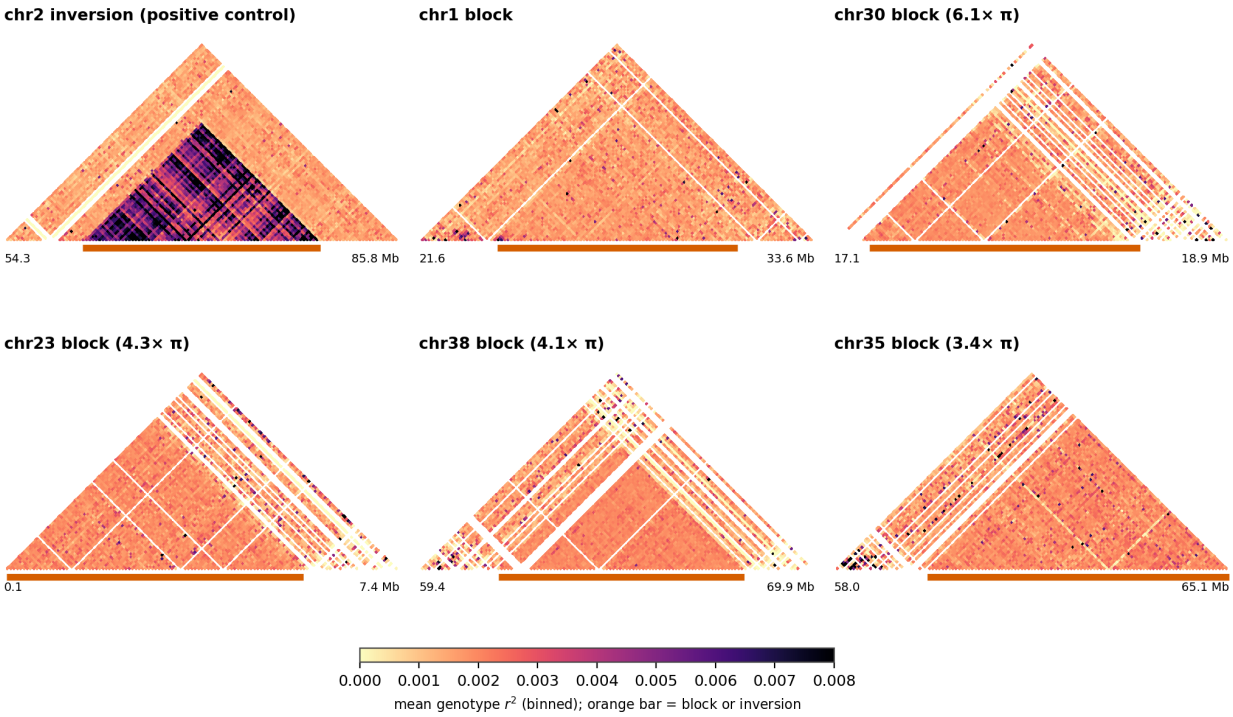


Figure S24: **The elevated-diversity blocks carry no inversion-like linkage disequilibrium.** Genotype r^2 among common SNPs (minor-allele frequency $\geq$0.1, $\leq$30% missing, 1,500 SNPs per region), averaged in 120 bins, for the chromosome-2 inversion (positive control) and five of the strongest blocks, each shown with a third of its length of flanking sequence. The orange bar marks the inversion or block. The inversion is a solid block of long-range LD over exactly its span (mean r^2 at >1 Mb 0.0056 inside vs 0.0016 in its flanks). None of the blocks differs from its flanks (0.0017–0.0020 vs 0.0012–0.0021 at >1 Mb, across all eight blocks tested), so none behaves as a segregating inversion or other long non-recombining haplotype. White stripes are bins with too few sampled SNPs.

Supplementary tables

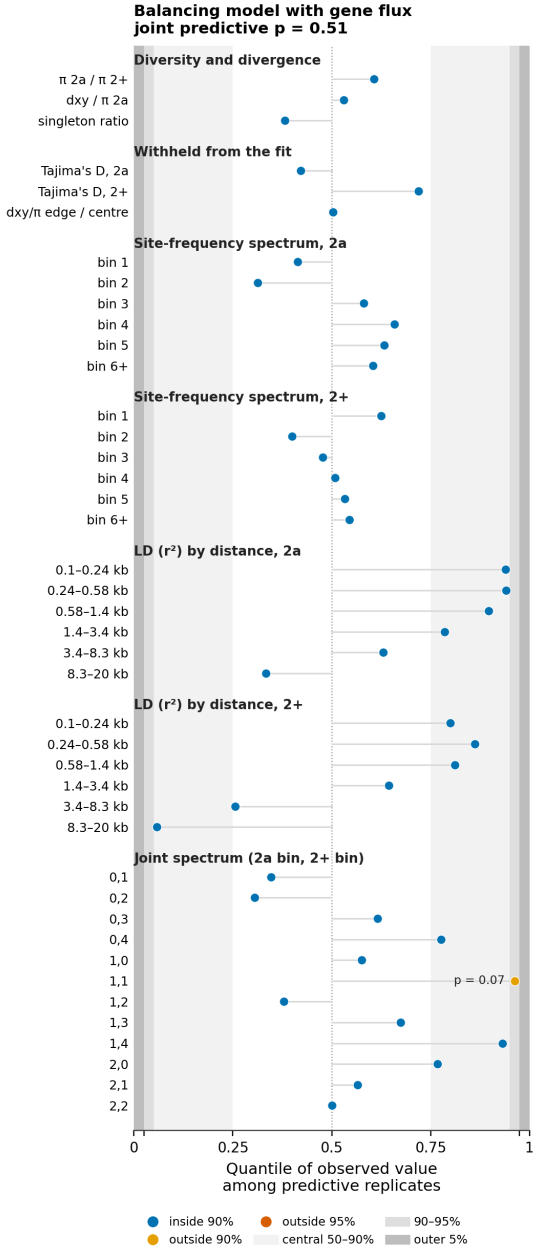


Figure S25: **Posterior predictive check of the balancing model with gene flux.** 1,000 NPE posterior draws were simulated again, and each point is the quantile of the observed statistic among the replicates. The shading marks the central 50–90%, the 90–95% and the outer 5% of the predictive distribution, the points are coloured by the two-sided predictive p (below 0.05, 0.05–0.10, above 0.10), and statistics with p below 0.10 are labelled. The title gives the joint predictive p. Tajima's D and the edge-to-centre divergence ratio were withheld from the fit. The site-frequency-spectrum bins are the proportions of segregating sites with 1 to 5 and 6 or more minor-allele copies of 20, and the joint-spectrum cells give the count bins of 2a and 2+ (0, 1–4, 5–15, 16–19 and 20 of 20). The check of the model without gene flux is summarised in the Supplementary Methods.

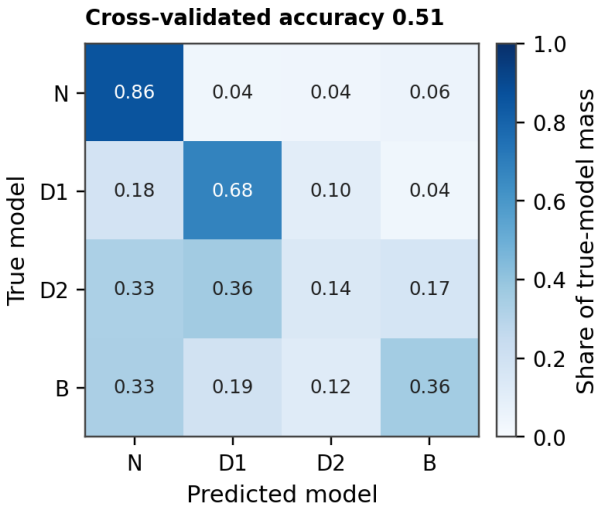


Figure S26: **Confusion matrix of the ABC-RF model choice for the chromosome-2 inversion.** Grouped five-fold cross-validation over 16,959 simulations of the four histories with gene flux, neutral drift (N), directional selection still under way (D1) or stopped (D2) and balancing selection (B). Each row is the true history, and the cells give the share of its prior-weighted mass assigned to each predicted history, so rows sum to 1.

Table S1: **Software and versions.** Versions were read off the analysis machine (conda metadata, `pip list`, `--version`, or recorded `versions.txt`).

Tool (version)	Used for	Reference
msinv 0.4.1	chr2 inversion simulation (ABC-RF, NPE)	(Small, 2025; Kelleher et al., 2018)
grenepipe	read processing and variant calling	(Czech and Exposito-Alonso, 2022)
fastp	read trimming	(Chen et al., 2018)
BWA-MEM2	read mapping	(Vasimuddin et al., 2019)
Picard MarkDuplicates	duplicate removal	Broad Institute
GATK HaplotypeCaller	joint genotyping (gVCF, all sites)	(Poplin et al., 2017)
GenMap 1.3.0	mappability (150-mers, 2 mismatches)	(Pockrandt et al., 2020)
plink2	per-division allele counts (per-SNP Hudson F_{ST})	(Chang et al., 2015)
LiftOn	annotation transfer to *S. oualaniensis*	(Chao et al., 2025)
MACSE 2.07	codon alignment (sex-biased mutation)	(Ranwez et al., 2018)
EnTAP	functional annotation	(Hart et al., 2020)
ANGSD 0.940	SFS, θ, diversity (demography, diversity)	(Korneliussen et al., 2014)
Stairway Plot 2 (2.1.1)	demography	(Liu and Fu, 2020)
GONE2 1.0.4	recent demography	(Santiago et al., 2025)
moments 1.6.0	demographic inference (demography)	(Jouganous et al., 2017)
demes 0.2.3	demography interchange	(Gower et al., 2022)
ReLERNN 0.2	recombination map	(Adrion et al., 2020)
diploSHIC 1.1.0	sweep scan	(Kern and Schrider, 2018)
RAiSD (`0c705a8`)	sweep corroboration (sweep scan)	(Alachiotis and Pavlidis, 2018)
SLiM 5.2	forward simulation	(Haller et al., 2026)
msprime 1.4.2	coalescent sim, mutation overlay	(Baumdicker et al., 2022)
tskit 1.0.3	tree-sequence I/O	(Kelleher et al., 2018)
discoal 0.1.7a	sweep-sim validation	(Kern and Schrider, 2016)
pg_gpu 0.1.1.dev14	windowed $\pi/d_{xy}/F_{ST}$, PCA	(Pope et al., 2026)
mkado 0.5.0	McDonald–Kreitman	(Rivera-Colón et al., 2026)
DIAMOND 2.1.12	homology (inversion, sweep scan)	(Buchfink et al., 2021)
HMMER 3.4	Pfam domain screen	(Eddy, 2011)
Pfam-A	domain screen	(Mistry et al., 2021)
AnchorWave 1.2.3	*I. coindetii*↔*Illex illecebrosus* alignment	(Song et al., 2022)
minimap2 2.29	alignment	(Li, 2018)
degenotate 1.3	0-/4-fold degeneracy	(Thomas, 2024)
bedtools 2.30.0	interval operations	(Quinlan and Hall, 2010)
bcftools/samtools 1.21	VCF/BAM handling	(Danecek et al., 2021)
EarlGrey 7.0.1	TE annotation	(Baril et al., 2024)
StringTie 3.0.0	transcript assembly	(Kovaka et al., 2019)
SQANTI3	Iso-Seq QC	(Pardo-Palacios et al., 2024)

Table S2: **Chromosome-2 karyotype counts and derived-arrangement frequency by NAFO division** (ordered by latitude). The derived (2a) frequency p_{2a} shows no latitudinal cline (range 0.50–0.70, excluding 3N with one individual), consistent with geographic $F_{ST} \approx 0$ (fig. S6).

Division	2a/2a	2a/2+	2+/2+	N	p_{2a}	mean lat. (°N)
6C	33	44	9	86	0.640	36.9
6B	32	49	16	97	0.582	38.3
6A	32	47	15	94	0.590	39.2
3N	0	0	1	1	0.000	43.1
4X	31	41	11	83	0.620	43.2
4W	19	19	9	47	0.606	43.6
3O	4	9	3	16	0.531	44.0
4T	6	11	6	23	0.500	47.7
4R	26	26	4	56	0.696	48.7
4S	7	5	3	15	0.633	49.9
3K	33	48	20	101	0.564	50.0
Total	223	299	97	619	0.602	—

Table S3: **Use of the outgroups.** *I. coindetii* served three roles, the structural comparison of the chromosome-2 arrangements, a count of allele sharing at sites that differ between them, and the divergence term of the genome-wide McDonald–Kreitman (MK) test, the latter resting on a whole-genome *I. coindetii*↔*Illex illecebrosus* AnchorWave alignment. *I. argentinus* was aligned identically (CDS) and used as a secondary MK outgroup, and its reads were counted at the sites that differ between the arrangements. It yields ∼9× fewer fixed coding differences than *I. coindetii* yet has genome-wide π indistinguishable from *Illex illecebrosus*, marking it as a recently-diverged congener of near-identical diversity. *S. oualaniensis* was used only for the structural comparison at the breakpoints, with positions given on the primary assembly and on HiC_scaffold_1.

Step	Quantity	Value
Alignment (AnchorWave v1.2.3, CDS-anchored)		
	I. coindetii substitution sites (genome-wide)	45,722,335
	aligned scaffold identity (inversion)	99.2–99.9%
Chromosome-2 arrangements in the reference individual (minimap2)		
	hap1 vs. primary, inverted segment	60.54–79.53 Mb
	hap2 vs. primary	collinear
	arrangement of hap1 / hap2	2a / 2+
	I. coindetii 37.95–74.95 Mb, aligned to 2+	one inverted run
	… aligned to 2a	two runs, split at 60.5 Mb
	I. coindetii adjacency at 56.9 Mb	kept in 2+, broken in 2a
Chromosome-2 breakpoints in *S. oualaniensis* (HiC_scaffold_1, minimap2)		
	Illex illecebrosus chr2 genes transferred to HiC_scaffold_1	265 of 319
	left breakpoint, genes at 60.49 and 60.55 Mb	adjacent (51.52, 51.57 Mb)
	right breakpoint, 78.2–83.6 Mb	one inverted run
	2+ adjacencies (hap2), both breakpoints	present
	2a adjacencies (hap1), both breakpoints	absent
Allele sharing with *I. coindetii* (inversion, frequency difference between 2a and 2+ ≥ 0.7)		
	differentiated sites (*I. coindetii* base callable)	15,093 (14,650)
	… 2a majority allele = *I. coindetii*	26.0%
	I. coindetii differs from the reference base	4,582
	… 2a / 2+ carries the *I. coindetii* allele	65.0% / 35.0%
I. argentinus reads at the same sites (10 individuals, ∼0.3× each)		
	sites covered / reads	13,195 / 34,860
	reference bias, collinear control	$0.018 + 1.02p$
	2a frequency, pooled (per individual)	0.284 (0.27–0.30)
	sites with ≥4 reads showing one allele	87%
	2a allele where *I. coindetii* carries 2a / 2+	0.68 / 0.24
McDonald–Kreitman test (mkado, *I. coindetii* divergence)		
	genes tested	20,999
	nonsynonymous / synonymous divergence (Dn / Ds)	15,343 / 13,670
	nonsynonymous / synonymous polymorphism (Pn / Ps)	468,124 / 630,949
	standard α, pooled counts	0.339
	asymptotic α (95% bootstrap CI)	0.642 (0.630–0.656)
I. argentinus (secondary MK outgroup, CDS alignment)		
	nonsynonymous / synonymous divergence (Dn / Ds)	1,355 / 1,805
	fixed coding differences (vs *I. coindetii* 29,013)	3,160 (∼9× fewer)
	asymptotic α	0.526
	genome-wide π (ANGSD, 6 autosomes; *Illex illecebrosus* = 0.0093)	0.0092

Table S4: **Gene-ontology enrichment of the Tier-1 (diploSHIC-only, Markov-corrected) sweep genes.** The 623 Tier-1 calls overlap 885 genes (475 GO-annotated). Enrichment by Fisher's exact test against the annotated-gene background on the scanned chromosomes, Benjamini–Hochberg FDR. 124 biological-process terms reach FDR<0.05, and representative terms are shown, spanning the two dominant themes (neural development and RNA-regulatory or transcriptional control). All are carried by the hard-sweep calls. The soft-sweep gene set has no term at FDR<0.05. The full machine-readable table is `hmm_decode_masked/tier1_GO_enrichment.tsv`.

GO id	term (biological process)	genes (T1/bg)	fold	FDR
GO:0048699	generation of neurons	123/1441	1.59	2.0×10^{-4}
GO:0022008	neurogenesis	125/1508	1.54	2.6×10^{-4}
GO:0030182	neuron differentiation	91/1075	1.57	1.6×10^{-3}
GO:0030154	cell differentiation	186/2548	1.36	5.4×10^{-4}
GO:0009887	animal organ morphogenesis	96/1115	1.60	8.9×10^{-4}
GO:0050767	regulation of neurogenesis	66/693	1.77	1.3×10^{-3}
GO:0000902	cell morphogenesis	72/775	1.73	1.3×10^{-3}
GO:0007409	axonogenesis	45/463	1.81	1.0×10^{-2}
GO:0048858	cell projection morphogenesis	59/609	1.80	1.6×10^{-3}
GO:0051252	regulation of RNA metabolic process	145/1920	1.40	1.3×10^{-3}
GO:2001141	regulation of RNA biosynthetic process	135/1768	1.42	1.6×10^{-3}

Table S5: **Functional clusters of the background-selection-robust Tier-1 sweep genes.** The 151 GO terms enriched at FDR<0.05 in the BGS-robust set (137 biological process, 12 molecular function and 2 cellular component, fig. S15), each assigned to one functional cluster by keyword rules on the term name (`scripts/tableS5_go_clusters.py`). Transcriptional regulation and neural development are the largest clusters, followed by organ development and cell morphogenesis/differentiation, with a distinct photoreception/sensory cluster. Fold range and the representative term (lowest FDR) are given per cluster.

functional cluster	terms	fold range	representative term (min FDR)
RNA / transcription regulation	36	1.4–2.7	pos. reg. nucleobase-containing compound metabolic process (9×10^{-4})
Neural development	28	1.6–22.2	generation of neurons (3×10^{-5})
Organ / anatomical development	21	1.2–2.5	anatomical structure morphogenesis (2×10^{-3})
Cell morphogenesis & differentiation	13	1.5–2.2	cell development (2×10^{-5})
Photoreception / sensory response	7	1.5–13.8	response to light stimulus (9×10^{-5})
Other regulation / metabolism	46	1.2–11.4	reg. response to stimulus (3×10^{-3})

Table S6: **Tier-2 (cross-method-concordant) sweep genes.** The 24 genome-wide-scan regions where the diploSHIC-HMM sweep signal is independently corroborated by an LD/SFS-footprint method (RAiSD or SweepFinder2). Ten regions carry a named protein-coding gene (13 genes, listed). The remaining concordant regions are gene deserts or uncharacterised loci. At the two regions marked † the gene is the target assigned by fine-scale localisation of the diversity minimum, as in main-text Table 4. Coordinates are in Mb, and α and DoS are from the McDonald–Kreitman test against *I. coindetii* ('–' where the test is unpowered, i.e. too few fixed/polymorphic sites). The full candidate set, comprising all 227 diploSHIC-outlier regions, the 24 concordant regions and the per-region background-selection test, is provided as machine-readable tables (`outlier_scan_45_masked/{windows,regions}.tsv` and the `concordant` gene/MK/BGS tables).

region (Mb)	sweep	gene	EnTAP best hit	α	DoS
1:20.3	hard	*Aplnr*	Apelin receptor	1.00	+0.17
1:85.5	hard	*Cpes* †	Ceramide phosphoethanolamine synthase	–	-0.52
2:84.8	hard	*NDRG3*	Protein NDRG3	–	-0.18
5:38.0	hard	*cept1*	Choline/ethanolaminephosphotransferase 1	–	-1.00
5:38.0	hard	*Arsi*	Arylsulfatase I	–	-0.25
6:3.0	hard	*Rexo1*	RNA exonuclease 1 homolog	–	-0.53
13:73.2	hard	*Rab3*	Ras-related protein Rab-3	–	-0.17
13:73.2	hard	*hyi*	Putative hydroxypyruvate isomerase	–	-0.64
13:73.2	hard	*Vps28*	Vacuolar protein sorting-associated protein 28	–	–
25:65.0	soft	*GALT* †	Galactose-1-phosphate uridylyltransferase	0.07	+0.02
32:9.3	soft	*MACROD2*	ADP-ribose glycohydrolase MACROD2	–	-0.59
33:9.0	hard	*Tinagl1*	Tubulointerstitial nephritis antigen-like	–	-0.23
35:39.0	hard	*ZEB2*	Zinc finger E-box-binding homeobox (ZEB)	–	-0.31

Table S7: **Genes captured by the chromosome-2 inversion** (chr2:60.54–79.50 Mb). The 45 functionally annotated genes (of 89 gene models), ordered by position. The column fixed is the number of exonic fixed differences between the 2a and 2+ arrangements ($|\Delta AF| \geq 0.9$, '–' for none). The column sums to 24 of the 26 exonic fixed differences, and the other two fall in the unannotated gene model `00005332`. EnTAP best-hit descriptions. The set spans lipid metabolism (phospholipases, PNPLA, malonyl-/acyl-CoA transferases), peroxisomal and mitochondrial function, DNA repair (XRCC1, TDG, WRNIP1), transcription/splicing (ERG, PRPF8, WDR82) and signalling (PRKCD), with no coherent pathway, consistent with the absent GO enrichment (fig. S21). Gene IDs are the `LOC_` suffix.

region (Mb)	gene	EnTAP best hit	fixed
2:60.6	00005299	protein HGH1 homolog isoform X1	–
2:60.9	00005300	Phospholipase A2 domain-containing protein	3
2:61.4	00005301	lon protease homolog 2, peroxisomal isoform X2	–
2:61.4	00005302	39S ribosomal protein L18, mitochondrial	–
2:61.4	00005303	sodium- and chloride-dependent taurine transporter	–
2:61.6	00005304	cytochrome P450 4X1-like	5
2:61.6	00005305	uncharacterized protein LOC115219520	–
2:64.1	00005315	PNPLA domain-containing protein	–
2:64.1	00005316	PNPLA domain-containing protein	–
2:65.5	00005319	CCHC-type domain-containing protein	–
2:65.7	00005320	uncharacterized protein LOC127831186	–
2:66.1	00005322	solute carrier family 66 member 2 isoform X2	–
2:66.3	00005324	Ubiquitin-conjugating enzyme E2 T	–
2:68.0	00005330	PCDHD1 (protocadherin)	–
2:68.8	00005334	retrovirus-related Pol polyprotein (transposon)	–
2:69.0	00005337	PNPLA domain-containing protein	–
2:69.0	00005338	ELMO domain-containing protein	–
2:69.1	00005339	malonyl-CoA-acyl carrier protein transacylase, mito.	1
2:69.1	00005340	acidic repeat-containing protein-like	6
2:69.1	00005341	RPGRIP1L (ciliary)	2
2:70.2	00005347	transcriptional regulator ERG homolog	–
2:71.2	00005349	transcriptional regulator ERG homolog	–
2:71.6	00005350	uncharacterized protein LOC115215517	3
2:71.9	00005351	cysteine-rich venom protein	2
2:72.0	00005352	cysteine-rich venom protein	–
2:74.6	00005356	DUF7043 domain-containing protein	–
2:77.7	00005360	SWIM-type domain-containing protein	–
2:77.8	00005361	UPB1 (β-ureidopropionase)	–
2:77.8	00005362	DNA-repair protein XRCC1 (N-terminal domain)	–
2:77.8	00005363	TIM21 (mitochondrial import)	–
2:77.9	00005364	dynein light chain roadblock-type 2	–
2:77.9	00005365	TDG (thymine-DNA glycosylase)	–
2:78.0	00005367	small integral membrane protein 4	–
2:78.1	00005368	uncharacterized protein LOC115226124	–
2:78.1	00005369	ATPase WRNIP1-like	–
2:78.2	00005371	unnamed protein product (*Sepia*)	1
2:78.2	00005372	PPM1H (protein phosphatase)	–
2:78.2	00005373	WDR82	–
2:78.2	00005374	glycerate kinase-like	–
2:78.4	00005375	PRKCD (protein kinase C δ)	–
2:78.6	00005376	acyl-coenzyme A amino acid N-acyltransferase 1-like	–

region (Mb)	gene	EnTAP best hit	fixed
2:78.7	00005378	uncharacterized protein LOC115226077	–
2:78.7	00005379	PRPF8 (pre-mRNA splicing factor)	1
2:78.8	00005380	synaptic vesicle 2-related protein-like	–
2:79.3	00005384	unnamed protein product (*Sepia*)	–

Table S8: **Sampling events.** Every collection event (NAFO division, date, depth) with its source, the number of individuals analysed in this study (619 of the 633 whole-genome sequenced individuals, excluding 12 whose BAMs contain essentially no data and two likely contaminated individuals) and the number retained by Baker et al. (2025) after their genotype-missingness filter (540). Divisions are ordered north to south by mean sampling latitude. 4X was bulk-sampled across the division with no per-tow depth or position recorded.

Division	Date	Source	Depth (m)	N (this study)	N (Baker)
3K	16 August 2023	Recreational jig	3	48	48
	30 August 2023	Coastal fishery jig	5	53	46
4S	28 August 2023	Canadian trawl survey	228	2	2
	29 August 2023	Canadian trawl survey	251	1	1
	30 August 2023	Canadian trawl survey	80	1	1
	30 August 2023	Canadian trawl survey	124	2	2
	31 August 2023	Canadian trawl survey	152	2	1
	31 August 2023	Canadian trawl survey	155	3	3
	1 September 2023	Canadian trawl survey	273	1	1
	1 September 2023	Canadian trawl survey	296	1	1
	3 September 2023	Canadian trawl survey	243	2	2
4R	7 August 2023	Canadian trawl survey	113	36	24
	25 August 2023	Canadian trawl survey	173	2	2
	25 August 2023	Canadian trawl survey	233	5	4
	25 August 2023	Canadian trawl survey	245	9	7
	28 August 2023	Canadian trawl survey	221	4	3
4T	2 September 2023	Canadian trawl survey	243	1	0
	7 September 2023	Canadian trawl survey	27	1	1
	8 September 2023	Canadian trawl survey	49	1	0
	9 September 2023	Canadian trawl survey	112	1	0
	9 September 2023	Canadian trawl survey	183	1	0
	9 September 2023	Canadian trawl survey	306	1	0
	10 September 2023	Canadian trawl survey	81	1	1
	10 September 2023	Canadian trawl survey	101	1	1
	10 September 2023	Canadian trawl survey	285	1	0
	12 September 2023	Canadian trawl survey	285	1	0
	15 September 2023	Canadian trawl survey	50	1	1
	22 September 2023	Canadian trawl survey	39	1	1

Division	Date	Source	Depth (m)	N (this study)	N (Baker)
	22 September 2023	Canadian trawl survey	142	2	1
	22 September 2023	Canadian trawl survey	158	1	1
	23 September 2023	Canadian trawl survey	341	2	2
	24 September 2023	Canadian trawl survey	118	1	0
	27 September 2023	Canadian trawl survey	36	1	0
	28 September 2023	Canadian trawl survey	56	1	1
	28 September 2023	Canadian trawl survey	62	1	1
	29 September 2023	Canadian trawl survey	49	2	1
3O	13 October 2023	Canadian trawl survey	204	14	7
	14 October 2023	Canadian trawl survey	350	1	1
	14 October 2023	Canadian trawl survey	613	1	1
4W	16 July 2023	Canadian trawl survey	84	47	43
4X	July 2022	Canadian trawl survey	not recorded	83	76
3N	20 October 2023	Canadian trawl survey	160	1	1
6A	22 September 2022	US trawl survey	120	94	84
6B	18 June 2023	Commercial trawl	74	49	46
	16 August 2023	Commercial trawl	201	48	48
6C	14 September 2022	US trawl survey	307	86	73
Total				619	540

Table S9: **Anomalous-diversity blocks, masked from the sweep scan and gene-set analyses.** Blocks of ≥1 Mb whose rolling nucleotide diversity exceeds twice the chromosome median. π fold is the block mean over the chromosome median. Division ratio is the range of block-to-flank diversity across the ten sampling divisions (ANGSD). Depth is block-to-flank read depth, as the median over the 621 individuals with data (Illumina, mapping quality ≥0) and for the assembly individual (PacBio HiFi, mapping quality ≥20). Non-unique is the percentage of the block's accessible sequence that 150-bp reads cannot map uniquely (GenMap, up to two mismatches), against 3.8% genome-wide. Class is collapsed duplication (population depth ≥1.15), under-mapped (depth ≤0.90), or elevated diversity with normal depth. In long-read genotypes of two individuals, collapsed blocks carry five to six times the flanking density of heterozygous sites at skewed allele fractions (median minor-allele fraction 0.29–0.33 vs 0.38–0.42 in flanks), and elevated-diversity blocks carry about twice the flanking density at normal fractions (chr1, 0.41 vs 0.40–0.43). The under-mapped chr21:0.0–9.4 block also shows skewed fractions (0.25 and 0.28 vs 0.42 and 0.44), with heterozygous-site density halved in the assembly individual, whose second haplotype largely fails to map, and tripled in the second individual, consistent with under-mapped blocks being largely duplicated within the assembly (33% of their accessible sequence non-unique).

Block	Mb	π fold	D block / chrom	Genes/Mb	Division ratio	Depth pop. / HiFi	Non-unique (%)	Class
chr1:24.0–31.3	7.3	2.8	−1.77 / −2.21	1.8	2.20–2.37	0.97 / 1.05	7	elevated diversity
chr6:12.0–18.9	6.9	2.3	−1.58 / −2.17	1.7	1.26–1.34	0.62 / 0.92	45	under-mapped
chr6:19.3–23.2	3.9	2.4	−1.56 / −2.17	1.8	1.27–1.32	0.66 / 0.84	41	under-mapped
chr8:29.7–35.5	5.8	2.3	−1.97 / −2.22	1.7	1.97–2.12	0.98 / 1.10	7	elevated diversity
chr9:27.2–29.9	2.7	3.6	−1.12 / −2.12	2.2	3.01–3.19	1.19 / 1.18	9	collapsed duplication
chr10:66.7–69.4	2.7	2.1	−2.06 / −2.26	0.4	1.46–1.53	0.95 / 1.01	2	elevated diversity
chr11:44.1–45.5	1.4	2.1	−1.83 / −2.15	2.1	1.48–1.53	0.97 / 1.04	3	elevated diversity
chr13:33.5–39.0	5.5	2.7	−1.79 / −2.19	3.1	2.50–2.79	0.93 / 1.05	4	elevated diversity
chr21:0.0–9.4	9.4	2.6	−1.39 / −1.94	1.7	2.22–2.49	0.68 / 0.61	36	under-mapped
chr21:74.4–79.4	5.0	2.5	−1.42 / −1.94	2.0	2.09–2.30	0.66 / 0.61	39	under-mapped
chr23:0.1–5.6	5.5	4.3	−0.96 / −2.04	2.2	3.94–4.43	1.78 / 1.91	12	collapsed duplication
chr23:62.4–65.3	2.9	2.3	−1.68 / −2.04	2.8	1.28–1.33	0.88 / 1.11	8	under-mapped
chr23:65.7–67.7	2.0	2.3	−1.73 / −2.04	1.5	1.23–1.27	1.07 / 1.12	10	elevated diversity
chr25:61.4–63.8	2.4	2.8	−1.57 / −2.17	2.9	2.68–2.86	1.16 / 1.16	25	collapsed duplication
chr29:34.4–40.7	6.3	2.6	−1.41 / −1.97	2.1	2.44–2.79	0.86 / 0.98	30	under-mapped
chr30:17.3–18.5	1.2	6.1	−0.69 / −2.17	2.5	5.77–6.17	2.19 / 2.66	2	collapsed duplication
chr34:29.9–44.7	14.8	2.6	−1.45 / −1.95	1.8	2.20–2.43	0.81 / 0.93	33	under-mapped
chr35:59.7–65.1	5.4	3.4	−1.47 / −2.13	1.9	3.21–3.52	1.53 / 1.84	4	collapsed duplication
chr38:19.0–31.8	12.8	2.8	−1.41 / −1.97	2.2	2.70–3.08	0.87 / 1.17	30	under-mapped
chr38:61.5–68.0	6.5	4.1	−1.08 / −1.97	2.6	4.11–5.08	1.44 / 1.21	17	collapsed duplication
chr43:45.7–61.6	15.9	2.4	−1.52 / −1.84	2.2	1.83–2.02	0.93 / 0.94	31	elevated diversity
chr44:8.8–11.7	2.9	2.7	−1.27 / −1.85	1.4	2.35–2.59	0.96 / 0.88	5	elevated diversity

Table S10: **Z recombination and sex-biased mutation.** Top, the ratio of population-scaled recombination rates on the Z and the autosomes in the 329 ZZ males from the decay of σ_D^2 with distance (86 autosomal and nine Z windows of 2 Mb, 95% CI from 200 bootstrap resamples of windows). Bottom, the ratio R of Z to autosomal synonymous divergence between *Illex illecebrosus* and *S. oualaniensis* and the male-to-female mutation ratio $\alpha = (3R-2)/(4-3R)$, following Miyata et al. (1987), for all genes and with autosomal genes matched to the Z genes by d_N/d_S, by third-position GC content (GC3) or by both, or restricted to genes above the autosomal median d_N/d_S (95% CI from 2,000 bootstrap resamples of genes within strata).

Linkage disequilibrium		ρ_Z/ρ_A	95% CI
Ohta–Kimura decay fit		1.04	0.88–1.28
Curve rescaling		1.08	0.78–1.67
Synonymous divergence	**Z / autosomal genes**	**R (95% CI)**	**α (95% CI)**
All genes	109 / 8,649	0.83 (0.72–0.94)	0.33 (0.10–0.70)
Matched on d_N/d_S	109 / 8,649	0.84 (0.73–0.96)	0.36 (0.11–0.79)
Above median d_N/d_S	44 / 4,324	0.79 (0.66–0.93)	0.24 (−0.01–0.65)
Matched on GC3	109 / 8,649	0.87 (0.76–0.98)	0.43 (0.17–0.89)
Matched on d_N/d_S and GC3	109 / 6,042	0.89 (0.78–1.01)	0.49 (0.20–1.06)